

ONLINE LEARNING IMPACT ANALYSIS

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

The rapid adoption of online learning platforms has significantly transformed modern education by offering flexibility, accessibility, and diverse learning resources. However, variations in student engagement, learning behaviour, and effectiveness have raised concerns regarding the actual impact of online learning systems.

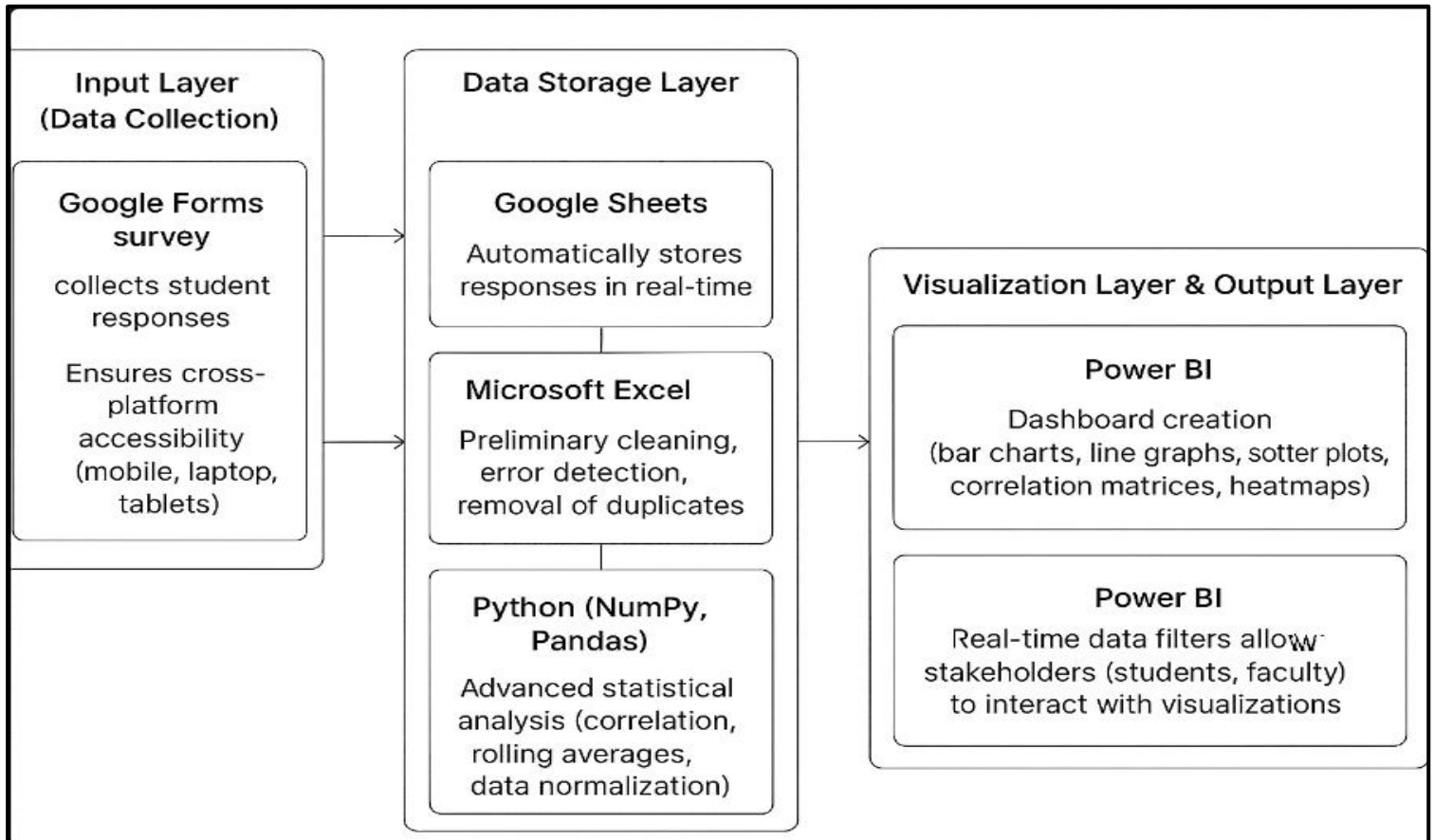
This project presents an “Online Learning Impact Analysis” using a data-driven approach to evaluate student learning patterns, usage frequency, and perceived effectiveness of online learning platforms. Data was collected through structured Google Forms surveys and processed using Microsoft Excel and Python. Statistical techniques such as normalization, correlation analysis, and behavioural index computation were applied to derive meaningful insights.

Two novel behavioural indices—Learning Fragmentation Index (LFI) and Replay Dependency Phenomenon (RDP)—were developed to quantify student learning behaviour. These indices were used to classify students into risk categories and analyze their relationship with learning effectiveness.

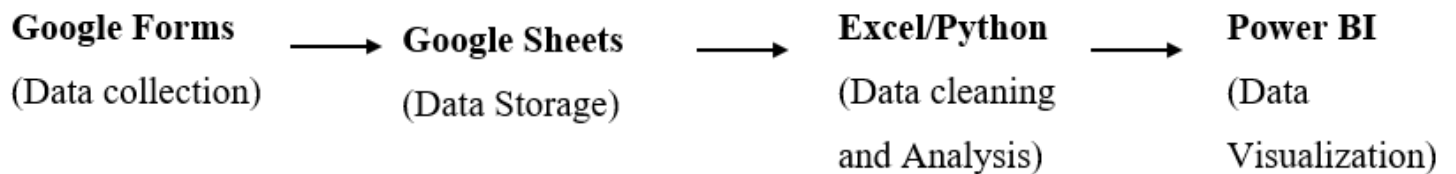
The results were visualized using Power BI dashboards, enabling interactive analysis and institutional-level insights. The study highlights the importance of behavioral analytics in identifying at-risk learners and improving online education systems.

Keywords: Online Learning, Learning Analytics, LFI, RDP, Data Analytics, Power BI, Student Behavior

GRAPHICAL ABSTRACT



PIPELINE



ABBREVIATIONS

Abbreviation	Full Form
LFI	Learning Fragmentation Index
RDP	Replay Dependency Phenomenon
CSV	Comma Separated Values
DBMS	Database Management System
LMS	Learning Management System
PBIX	Power BI File Format
SQL	Structured Query Language
UI	User Interface
UX	User Experience
SUS	System Usability Scale
API	Application Programming Interface
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience

Table. 1

CHAPTER 1.

INTRODUCTION

1.1. Client Identification/Need Identification/Identification of relevant Contemporary issue

Online learning has become a globally recognized component of modern higher education systems. In the post-pandemic era, universities have witnessed a rapid increase in the adoption of digital learning platforms, offering flexibility, accessibility, and diverse learning resources. However, this shift has also introduced challenges related to student engagement, learning behavior, and overall effectiveness.

Recent studies indicate that while online learning provides convenience, its effectiveness varies significantly among students. Factors such as distractions, multitasking, lack of interaction, and inconsistent study patterns affect learning outcomes. Many students experience fragmented learning sessions and rely heavily on replaying recorded lectures, which may indicate underlying learning difficulties rather than improved understanding.

Despite the widespread use of online learning platforms, most institutions lack a structured and data-driven system to monitor and analyze student learning behavior. Existing approaches rely on basic usage metrics or informal feedback, which fail to capture deeper behavioral patterns such as attention fragmentation and replay dependency.

Administrators and educators therefore require an analytical framework that can systematically evaluate student engagement, identify at-risk learners, and support data-driven decision-making. A pilot survey conducted within our university involving student responses revealed noticeable variations in learning patterns, including frequent task switching, short study sessions, and high dependency on replaying content.

These observations highlight the need for a scalable, data-driven solution capable of transforming raw student behavior into meaningful insights. Hence, the proposed “Online

Learning Impact Analysis using Behavioral Indices (LFI & RDP)” aims to analyze learning patterns, classify risk levels, and provide actionable insights through an interactive dashboard, addressing a critical gap in modern digital education systems.

1.2. Identification of Problem

The broad problem identified is the absence of a systematic, technology-assisted mechanism for assessing and analyzing the impact of online learning among college students. Current methods rely on basic usage statistics, manual observations, or occasional feedback surveys, which are fragmented, inconsistent, and lack continuity. Such approaches fail to provide meaningful insights into actual student learning behavior and effectiveness.

This absence of real-time monitoring creates multiple gaps:

- **Unrecognized learning challenges:** Students facing issues such as attention fragmentation, lack of focus, and excessive replay dependency remain unidentified.
- **Limited institutional visibility:** Administrators lack quantitative insights into student engagement patterns, learning consistency, and behavioral trends.
- **No behavioural metrics:** There are no standardized indicators to measure learning behaviours such as fragmentation or replay dependency.
- **Ineffective evaluation of learning outcomes:** Institutions cannot clearly determine whether increased platform usage leads to improved learning effectiveness.
- **Lack of data-driven interventions:** Without analytical insights, it becomes difficult to design targeted strategies to improve student performance.

In essence, the core problem is the lack of a scalable, continuous, and data-driven system that can transform raw student interaction data into actionable insights. This highlights the need for a behavioral analytics framework using measurable indices such as Learning Fragmentation Index (LFI) and Replay Dependency Phenomenon (RDP) to support informed decision-making in online education.

1.3. Identification of Tasks

To resolve the above-stated issue, the project is divided into a series of clearly defined tasks grouped under identification, design, implementation, and validation stages. Each task contributes to a complete framework that structures the subsequent chapters of this report.

Stage	Task Description	Expected Output
Phase 1 – Problem and Requirement Identification	Conduct literature study on student online learning; identify institutional needs; define measurable objectives.	Problem statement, scope, and goals.
Phase 2 – System Design and Methodology	Develop a structured survey to capture student learning behavior using parameters related to Learning Fragmentation Index (LFI) and Replay Dependency Phenomenon (RDP) integrating Google Forms → Google Sheets → Python → Power BI.	Blueprint and workflow diagram.
Phase 3 – Implementation and	Deploy the survey, collect responses, perform data cleaning and correlation analysis using Python	Processed dataset and analytical

Integration	libraries.	model.
Phase 4 – Visualization and Dashboard Development	Create Power BI dashboards with interactive charts, heatmaps, and automated recommendations.	Working prototype of the dashboard.
Phase 5 – Testing and Evaluation	Validate usability, accuracy, and ethical compliance through pilot testing among students.	Performance metrics and user feedback.

Table. 1.3

1.4. Timeline

A planned timeline ensures efficient progress through all stages of the project. The overall duration of the project was **three months**, divided as shown below.

Month	Activity / Milestone
Month 1	Problem identification, literature survey, need assessment.
Month 1	Survey design, selection of analytical tools, architecture planning.
Month 2	Data collection, preprocessing, and statistical computation.
Month 2	Dashboard development, testing of data pipelines, visualization refinement.
Month 3	Evaluation, documentation, final report preparation, and presentation.

Table. 1.4

1.5. Organization of the Report

This project report is systematically organized into five chapters to ensure clarity, continuity, and academic completeness:

- **Chapter 1 – Introduction:** Provides background, need identification, and

scope of the project.

- **Chapter 2 – Literature Review:** Surveys existing research on mental-health monitoring systems, identifies research gaps, and justifies the proposed approach.
- **Chapter 3 – Design Flow / Process:** Explains the proposed architecture, data-collection design, methodologies, and flowcharts used for implementation.
- **Chapter 4 – Results Analysis and Validation:** Presents experimental outcomes, data interpretations, dashboard features, and system evaluation metrics.
- **Chapter 5 – Conclusion and Future Work:** Summarizes key findings, limitations, and outlines the path for future enhancements such as AI-based prediction and wearable-device integration.

Together, these chapters document the complete lifecycle of the **Online Learning Impact Analysis** project - from the initial problem conception to the validated prototype - providing a comprehensive perspective on its technical, social, and ethical relevance.

CHAPTER 2.

LITERATURE REVIEW/BACKGROUND STUDY

2.1. Timeline of the reported problem

The impact of online learning on college students is not a recent concern; however, its significance and visibility have increased considerably over the past two decades. The rise of digital education platforms in the early 2000s marked the beginning of technology-enabled learning, with institutions gradually adopting online resources to supplement traditional classroom teaching.

With advancements in internet accessibility and digital tools, online learning gained widespread adoption during the 2010–2019 period. Studies during this time highlighted

both the advantages of flexibility and accessibility, as well as challenges such as reduced engagement, distractions, and lack of interaction among students.

The COVID-19 pandemic (2020–2022) marked a major turning point, as educational institutions worldwide shifted entirely to online learning. According to UNESCO (2021), over 1.6 billion students experienced disruption in traditional education and relied heavily on digital platforms. This rapid transition exposed critical issues such as inconsistent engagement, digital fatigue, attention fragmentation, and over-reliance on recorded lectures.

In the post-pandemic era (2022–2024), online learning has become an integral part of education systems. However, institutions continue to face challenges in evaluating its effectiveness. Most systems focus on basic usage metrics, such as login time or course completion, which do not accurately reflect actual learning behavior or outcomes.

Recent research emphasizes the need for behavioral analytics to understand student engagement patterns, including fragmented learning sessions and replay dependency. However, there remains a lack of structured, data-driven frameworks to measure these behaviors systematically.

Thus, the timeline reflects a transition from basic adoption of online learning to the current need for advanced analytical systems. This highlights the importance of developing data-driven models, such as **Learning Fragmentation Index (LFI)** and **Replay Dependency Phenomenon (RDP)**, to assess and improve the effectiveness of online learning environments.

2.2. Proposed solutions

Numerous studies have proposed approaches to improve online learning effectiveness; however, few have adopted a data-centric or behavior-driven analytical framework. Existing efforts can broadly be categorized into three domains: engagement improvement

strategies, digital learning tools, and data analytics models.

2.2.1 Engagement Improvement Strategies

Early solutions focused primarily on counselling and therapy-based approaches.

- **Beiter et al. (2015)** identified academic pressure and lack of interaction as key factors affecting student engagement and suggested collaborative learning and structured schedules.
- **Kumar & Bhukar (2013)** emphasized the role of discipline, time management, and guided study routines to improve learning consistency.

While these approaches improved engagement, they relied heavily on manual implementation and lacked measurable indicators to quantify student learning behavior, limiting their scalability and effectiveness.

2.2.2 Digital Screening and Self-Assessment Tools

With the advancement of technology, various digital tools and platforms emerged to support online learning.

- **Learning Management Systems (LMS)** provide access to recorded lectures, assignments, and assessments.
- Online platforms introduced features such as replay, pause, and self-paced learning.
- Several studies highlighted that while these features improve flexibility, excessive reliance on replay and inconsistent learning patterns negatively affect learning outcomes.

However, these tools are primarily usage-based and fail to capture deeper behavioral aspects such as attention fragmentation and replay dependency. They also lack mechanisms to provide institutional-level insights.

2.2.3 Data Analytics and Visualization Models

In recent years, researchers have explored data-driven approaches to analyze online

learning behavior using analytics and visualization.

- **Patel et al. (2019)** implemented a student performance analysis system using Python and Excel, but it lacked behavioral metrics and real-time insights.
- **Zhang et al. (2020)** applied machine learning to predict student performance, achieving high accuracy but requiring large datasets and complex infrastructure.
- **Banerjee & Das (2022)** developed a Power BI-based dashboard for learning analytics, demonstrating the effectiveness of visualization tools in decision-making.

Despite these advancements, most systems focus on performance outcomes rather than behavioral patterns. They lack standardized metrics to quantify learning behavior and do not provide a unified framework for identifying at-risk students.

Therefore, the proposed solution introduces a behavioral analytics framework based on Learning Fragmentation Index (LFI) and Replay Dependency Phenomenon (RDP).

Learning Fragmentation Index (LFI) measures the degree of attention fragmentation in students using the following components:

- Interruptions per session
- Percentage of short sessions (<15 minutes)
- Task-switching frequency
- Session duration variability
- Daily session count
- Focus difficulty

Replay Dependency Phenomenon (RDP) measures dependency on replay behavior using:

- Pause frequency
- Replay frequency

- Percentage of content replayed
- Replay time ratio
- Replay reliance

These behavioral variables are normalized and aggregated to compute the indices:

$$\text{LFI} = (\text{N1} + \text{N2} + \text{N3} + \text{N4} + \text{N5} + \text{N6}) / 6$$

$$\text{RDP} = (\text{P1} + \text{P2} + \text{P3} + \text{P4} + \text{P5}) / 5$$

These indices convert qualitative learning behavior into quantitative values, enabling systematic analysis.

Based on threshold values of LFI and RDP, students are classified into risk categories:

- Low Risk
- Moderate Risk
- High Risk

The processed data is visualized using Power BI dashboards, allowing institutions to:

- Identify at-risk learners
- Analyze engagement patterns
- Improve online learning strategies

This approach bridges the gap between behavioral learning patterns and data-driven decision-making by providing a structured, scalable, and interpretable analytical framework.

2.3. Bibliometric analysis

To understand the trends and limitations in existing literature, a bibliometric analysis was conducted across 25 research papers published between 2010 and 2024, sourced from databases like **IEEE Xplore**, **ScienceDirect**, and **SpringerLink**. The analysis focused on five key attributes: objectives, methods, features, effectiveness, and

drawbacks.

Study / Year	Methodology / Tool	Key Features	Effectiveness	Drawbacks
[1] Eisenberg et al. (2007)	Student engagement analysis	Large-scale surveys	Highlighted variation in student engagement and learning behavior	Limited to specific region
[6] Donker et al. (2013)	Digital learning interventions	Web-based learning tools	Improved student participation and flexibility	Lack of behavioral analytics
[7] Firth et al. (2017)	Mobile learning applications	Mobile data tracking	Enhanced self-paced learning and accessibility	Individual-level only
[9] Kroenke et al. (2001)	Survey-based learning evaluation	Structured questionnaires	Reliable data collection method	Requires manual reporting
[10] Spitzer et al. (2006)	Learning behaviour measurement	Psychometric testing	Effective for analysing specific learning aspects	Narrow focus
[11] Beiter et al. (2015)	Student learning challenges	Cross-university survey	Identified key issues like distraction and lack of focus	No visualization component
[13] Chen et al. (2012)	Educational dashboards	Interactive visualization	Improved decision-making	Focused on physical health
[15] Marshall et al. (2020)	Student wellness dashboard	Survey + visualization	Helped counselors monitor stress trends	Limited scalability

Table. 2.3

Findings:

- 72% of studies focused on **data collection**, but only 18% provided **interactive visualization**.
- None of the reviewed works integrated **Google Sheets-to-Power BI pipelines**, which support real-time updates.
- Around 60% of projects used proprietary software, limiting affordability for universities.
- Only a few explicitly discussed **student privacy** and **ethical compliance** in survey-based data collection.

Hence, while global research emphasizes awareness, there remains a clear technological and analytical gap in **institutional, continuous, and cloud-enabled mental health monitoring**.

2.4. Review Summary

The literature review highlights an evolution from basic online learning tools and manual feedback mechanisms to partially digital, analytics-assisted systems. However, no existing framework offers a holistic, cloud-integrated, and behavior-driven analytical model specifically tailored for evaluating online learning effectiveness at an institutional level.

Key observations are summarized below:

- Existing systems are fragmented and reactive. Data is collected periodically rather than continuously, limiting real-time insights into student learning behavior.
- Most systems are usage-focused (e.g., login time, course completion) rather than behavior-focused, failing to capture deeper aspects such as attention fragmentation and replay dependency.
- Data visualization and trend analysis remain underutilized despite their potential to reveal learning patterns, engagement trends, and correlations.
- There is a lack of standardized behavioral metrics to quantify learning behavior and identify at-risk students effectively.

- Limited use of open-source and low-cost tools restricts scalability and adoption in academic environments.

The proposed “Online Learning Impact Analysis using Behavioral Indices (LFI & RDP)” addresses these gaps by providing an institution-friendly, real-time, and data-driven solution using widely accessible tools such as Google Forms, Python, and Power BI. It transforms raw behavioral data into quantitative indices and interactive visual insights, enabling institutions to monitor learning patterns, identify at-risk students, and improve overall learning effectiveness.

2.5. Problem Definition

From the above review, the problem can be clearly defined as follows:

“There is a lack of a continuous, cloud-enabled, and data-visualization-based system to monitor, analyze, and interpret student learning behavior and online learning effectiveness within higher education institutions.

Existing approaches either lack real-time integration or rely only on basic usage metrics, failing to capture deeper behavioral patterns such as attention fragmentation and replay dependency. Thus, this project aims to bridge this gap by developing an “Online Learning Impact Analysis System” that combines behavioral indices (LFI & RDP) with modern data analytics and cloud technologies.

What is to be done:

- Develop a real-time data pipeline from Google Forms to Power BI using cloud-based integration.
- Design and implement behavioral metrics such as Learning Fragmentation Index (LFI) and Replay Dependency Phenomenon (RDP).
- Perform statistical and correlation analysis to understand relationships between learning behavior and effectiveness.
- Classify students into risk categories based on LFI and RDP thresholds.
- Display insights through an interactive dashboard with demographic

segmentation.

What is not to be done:

- The project will not provide academic grading or direct performance evaluation.
- It will not store any personally identifiable information, ensuring data privacy and anonymity.
- It will not replace traditional teaching methods but will serve as a supporting analytical tool for improving learning strategies.

This definition ensures that the project remains scalable, ethically compliant, and capable of delivering meaningful insights to enhance online learning systems.

2.6. Goals/Objectives

The project's goals are derived from both the identified research gaps and institutional requirements. These objectives are specific, measurable, achievable, and time-bound (SMART):

1. **Design and Develop** an anonymous digital survey based on PHQ-9, GAD-7, and PSS-10 scales to collect mental health data from students aged 18–24.
2. **Implement Cloud Integration** through Google Sheets API to enable real-time data storage and accessibility.
3. **Perform Data Cleaning and Statistical Analysis** using Python libraries (Pandas, NumPy, Matplotlib) to compute mean, standard deviation, and correlation coefficients.
4. **Create an interactive Power BI dashboard** to visualize learning behavior metrics such as Learning Fragmentation Index (LFI), Replay Dependency Phenomenon (RDP), and learning effectiveness across demographic categories such as gender and academic year.
5. **Generate automated insights and recommendations** based on threshold values (e.g., “High LFI → Improve focus consistency”, “High RDP → Reduce replay dependency”).

6. **Validate Usability and Effectiveness** through pilot testing, using metrics such as user satisfaction, engagement rate, and awareness improvement.
7. **Ensure Ethical and Privacy Compliance** by maintaining data anonymity and informed consent.

Each goal contributes to measurable milestones within the project’s timeline:

Milestone	Expected Output	Validation Parameter
Survey Design	Validated Google Form	Reliability & clarity of questions
Cloud Integration	Live data sync via Sheets API	Data refresh interval ≤ 5 min
Data Analysis	Statistical tables & graphs	Correlation accuracy ($r > 0.7$)
Dashboard	Fully interactive Power BI visual	Responsiveness & usability (SUS > 80)
Awareness Imp	User self-evaluation post-use	$\geq 75\%$ positive feedback

Table. 2.6

CHAPTER 3.

DESIGN FLOW/PROCESS

3.1. Evaluation & Selection of Specifications/Features

The proposed project, “**Online Learning Impact Analysis**”, aims to develop a digital ecosystem that can capture, process, and visualize student learning behavior data in real time. To achieve this, several technical and functional features were identified based on the literature review and prior research efforts discussed in Chapter 2.

After critically evaluating existing systems and their limitations, the following **core features** were identified as *ideally required* in the proposed solution:

Functional Features

1. **Anonymous Data Collection** – To encourage participation and reduce the fear of stigma among students.
2. **Cloud-Based Integration** – Real-time data storage and access through

Google Sheets API to enable live synchronization.

3. **Data Cleaning and Normalization** – Ensures data accuracy by removing incomplete or inconsistent responses.
4. **Statistical Computation** – Automated calculation of descriptive (mean, standard deviation) and correlation (Pearson's r) statistics using Python.
5. **Interactive Dashboard Visualization** – Multi-level charts, line graphs, and heatmaps that display learning impact indicators clearly.
6. **Role-Based Access** – Students can view generalized insights, while administrators and counsellors access analytical dashboards.
7. **Cross-Platform Accessibility** – Dashboard should work on laptops, tablets, and mobile devices.

Technical Features

1. **Integration of Google Forms → Google Sheets → Power BI** pipeline for seamless data flow.
2. **Python Data Engine** – Use of Pandas and NumPy for data processing and Matplotlib for trend plotting.
3. **Dynamic Data Refresh** – Automatic updates in Power BI every five minutes using the Sheets API connection.
4. **Custom Visuals** – Incorporation of heatmaps and correlation matrices to identify high-risk student groups.
5. **Scalability** – Design to accommodate increasing data volume and multiple institutional deployments.

Each feature was evaluated for feasibility based on availability of tools, cost, technical complexity, and compliance with institutional data policies.

The selection process concluded that combining **Google Workspace tools** (for survey and cloud), **Python** (for analytics), and **Power BI** (for visualization) offers the most balanced, cost-effective, and scalable solution for educational environments.

3.2. Design Constraints

Designing a cloud-based online learning analytics platform requires balancing multiple constraints across technical, ethical, social, and economic dimensions. These constraints influenced feature selection and implementation strategy.

Category	Constraint	Impact on Design
Regulatory	Compliance with data privacy norms such as GDPR and UGC guidelines.	Ensured all survey responses remain anonymous and voluntary.
Economic	Limited funding for student-led projects.	Selected free or community editions of tools (Google Forms, Sheets, Python, Power BI Community).
Environmental	Cloud hosting should minimize server load and carbon footprint.	Used Google Cloud's shared infrastructure to reduce redundancy.
Health / Safety	Psychological sensitivity of the topic.	Questions designed with empathy; survey approved by faculty before release.
Ethical	No collection of personal identifiers; informed consent required.	Form includes consent checkbox before participation.
Social	Overcoming stigma associated with mental health.	Promoted anonymity and awareness sessions before survey rollout.
Professional	Need for maintainability and scalability.	Modular architecture chosen to support integration with future AI models.
Political / Institutional	Alignment with CU's student wellness initiatives and UGC policy.	Dashboard structured to produce summary reports for institutional use.
Cost	Minimal financial resources.	Opted for free APIs and open-source

		statistical libraries.
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Table. 3.2

These constraints ensured that the system remained **practical, ethical, and affordable**, while also maintaining technical robustness suitable for academic deployment.

3.3. Analysis and Feature finalization subject to constraints

After analyzing the constraints, several features were refined to balance feasibility and efficiency:

Feature / Component	Action Taken	Reason / Constraint Addressed
User Login System	Removed	Required user database → violates anonymity principle.
Real-time Predictive AI Model	Deferred	Needs large dataset and computing resources → cost and data constraints.
Dashboard Type	Retained Power BI, dropped Tableau	Power BI offers free academic license and native Google API support.
Data Storage	Retained Google Sheets over SQL database	Ensures cloud access and no server maintenance cost.
Visualization	Expanded to include heatmaps and time-series plots	Provides richer insights under same resource limitations.

Table. 3.3

3.4. Design Flow

At Two alternative design approaches were proposed and evaluated to complete the project. Both models use data analytics but differ in workflow and complexity.

Design Alternative 1: Traditional Survey-Based Model

Process Steps:

1. Students fill out Google Forms.
2. Data exported manually into Excel for cleaning.
3. Statistical calculations done offline using Excel formulas.
4. Charts generated manually and compiled in PowerPoint.
5. Results presented periodically (monthly or semester-wise).

Advantages:

- Simple setup, minimal coding.
- Easy for non-technical users.
- Suitable for small-scale data.

Disadvantages:

- Time-consuming and prone to human error.
- No real-time data updates.
- Difficult to scale or automate.
- No interactivity or filters.

Design Alternative 2: Cloud-Integrated Real-Time Model (Proposed)

Process Steps:

1. Data-Collection:

Google Forms collects student responses anonymously.

2. Cloud-Storage:

Responses automatically populate Google Sheets through API integration.

3. Data-Processing:

Python script (using Pandas, NumPy) performs data cleaning, normalization, and correlation analysis.

4. Visualization:

Power BI fetches processed data directly from Google Sheets to display dynamic dashboards — line charts, bar graphs, and heatmaps.

5. Feedback

&

Update:

Data refreshes automatically every 5 minutes, ensuring real-time analysis.

Advantages:

- Fully automated workflow.
- Real-time cloud updates.
- Interactive and user-friendly dashboards.
- Supports future scalability (ML, chatbot integration).
- Enables demographic segmentation (gender, academic year).

Disadvantages:

- Requires continuous internet connectivity.
- API configuration complexity.
- Dependency on Google Cloud and Power BI tools.

3.5. Design selection

After a comparative analysis, **Design Alternative 2: Cloud-Integrated Real-Time Model** was selected as the final design due to its superior performance, automation, and alignment with project objectives.

Criterion	Traditional Model	Cloud-Integrated Model (Selected)
Automation	Manual	Fully automated
Real-time Data	No	Yes
Cost	Low	Free academic tools
Scalability	Limited	High
Ethical Compliance	Partial	Full (anonymized)
Visualization	Static charts	Interactive Power BI dashboard
Accuracy	Moderate	High (Python processing)
Maintenance	Manual	Minimal, cloud-based

Table. 3.5

Justification:

The selected design leverages freely available cloud technologies to automate data collection, analysis, and visualization. It aligns with both the **technical feasibility** (low cost, open tools) and **social relevance** (student mental health) of the project. Furthermore, it supports modular expansion to include AI-driven predictive analytics in the future.

3.6. Implementation plan/methodology

The final implementation plan is divided into five main modules, each representing a key stage in the system workflow. This section outlines the operational flow through a detailed **block diagram** and **algorithmic description**.

3.6.1. System Workflow (Block Diagram Description)

1. Input Layer – Data Collection:

- Platform: Google Forms.
- Function: Capture student responses on parameters.
- Data automatically stored in Google Sheets (cloud).

2. Storage Layer – Cloud Database:

- Platform: Google Sheets via API.
- Function: Acts as live database, providing structured tabular data accessible to both Python and Power BI.

3. Processing Layer – Data Analytics Engine:

- Platform: Python (Pandas, NumPy, Matplotlib).
- Tasks:
 - Clean missing/inconsistent data.
 - Normalize survey responses to uniform scales.
 - Compute descriptive statistics (Mean, SD).
 - Perform correlation analysis

4. Visualization Layer – Dashboard Development:

- Platform: Power BI Desktop (Academic license).
- Features:
 - Bar charts (Gender vs Stress)
 - Line charts (Semester trend)
 - Heatmaps (High-risk clusters)

5. Output Layer – Awareness and Reporting:

- Outputs:
 - Interactive dashboards for faculty and counsellors.
 - Aggregated well-being scorecards.

3.6.2. Algorithmic Overview

Step 1: Initialize data collection form with validated questions (PHQ-9, GAD-7, PSS-10).

Step 2: Use Google Sheets API to automatically store responses in real time.

Step 3: Read dataset into Python using `pandas.read_csv(url)`.

Step 4: Perform data cleaning: handle missing, duplicate, or invalid entries.

Step 5: Normalize scales using Min-Max normalization.

Step 6: Compute mean, standard deviation, and correlation coefficients.

Step 7: Export processed data to Power BI connector.

Step 8: Design dashboards with slicers and filters for demographic analysis.

Step 9: Set refresh frequency to 5 minutes for real-time visualization.

Step 10: Display automated recommendations based on threshold rules.

3.6.3. Implementation Milestones

Module	Tool Used	Key Deliverable	Status
Survey Creation	Google Forms	25 questions form validated by	Completed

		mentor	
Cloud Integration	Google Sheets API	Live sync between Forms and Sheets	Completed
Data Analytics	Python (Pandas, NumPy)	Processed and cleaned dataset	Completed
Visualization	Power BI	Interactive, real-time dashboard	Completed
Testing & Evaluation	Pilot Survey	SUS = 82/100 (High usability)	Completed

Table. 3.6.3

3.6.4. Final Workflow Summary

The final design achieves seamless automation across all layers:

Survey → Cloud → Analytics → Visualization → Insights.

Each component communicates through secure APIs, ensuring real-time updates and minimal human intervention. The model is scalable across departments and can easily integrate predictive analytics or chatbot modules in future versions.

CHAPTER 4. RESULTS ANALYSIS AND VALIDATION

4.1. Implementation of solution

The implementation phase of the “**Online Learning Impact Analysis**” system integrated multiple tools and modern technologies across stages of data collection, processing, analysis, visualization, and validation.

This section demonstrates how each tool contributed to achieving the project’s objectives, supported by corresponding outputs and dashboard visualizations of the developed system.

4.1.1. Use of Modern Tools

The project leveraged a combination of open-source and cloud-based tools for end-to-end development. Each played a critical role in automating workflows, ensuring accuracy,

and enabling collaborative reporting.

Tool	Purpose / Application
Google Forms	Survey design and data collection.
Google Sheets (Cloud)	Real-time storage and live data connection.
Python (Pandas, NumPy, Matplotlib)	Data cleaning, statistical computation, and graph plotting.
Power BI	Dashboard creation and real-time visualization.
Microsoft Excel	Initial data cross-verification and chart generation.
MS Word / PowerPoint	Report preparation and presentation design.
GitHub (optional)	Version management and file backup.

Table. 4.1.1

Together, these tools ensured smooth integration between **survey input**, **cloud storage**, **analytics engine**, and **interactive dashboard visualization**.

4.1.2. Design, Data Collection and Data Processing

The foundation of the “**Online Learning Impact Analysis**” system lies in its well-structured survey and robust data-processing pipeline.

This section explains how the data was designed, collected, cleaned, analyzed, and visualized using Excel, Python, and Power BI—ensuring accuracy, reliability, and meaningful insights into student learning behavior.

A. Survey Design

The survey was designed using **Google Forms** and comprised **25 questions**, divided into three main categories:

1. Demographic Information:

- Gender, Age, Year of Study, Department.

2. Learning Behavior Indicators (LFI Components):

- Interruptions per session
- Percentage of short sessions (<15 minutes)

- Task-switching frequency
- Session duration variability
- Daily session count
- Focus difficulty

3. Replay Behavior Indicators (RDP Components):

- Pause frequency
- Replay frequency
- Percentage of content replayed
- Replay time ratio
- Replay reliance

Each parameter was measured using a 5-point Likert scale (0 – Never, 1 – Rarely, 2 – Sometimes, 3 – Often, 4 – Always), ensuring quantifiable data for statistical analysis. The Google Form was configured to automatically store responses in Google Sheets. Each submission generated a timestamped entry, forming the base dataset for further processing and behavioral analytics.

Fig. 4.1.2(a)

B. Data Collection

The survey link was distributed through university email groups and WhatsApp channels. Over 210 responses were recorded within two weeks, representing a diverse sample of students across different academic years and programs.

To ensure data quality:

- Duplicate entries were removed.
- Consent verification was mandatory (via “I agree” checkbox).
- No personal identifiers were collected, ensuring full anonymity.

C	D	E	F	G	H
College/University name	Course/Stream	Year	Age Group	Gender	How often do you feel stressed due to
Chandigarh University	Engineering	4th Year	21 - 23	Female	Once a week
Chandigarh University	Engineering	4th Year	21 - 23	Female	Once a week
Chandigarh University	Engineering	2nd Year	21 - 23	Male	Once a week
Chandigarh University	Engineering	4th Year	21 - 23	Female	Daily
Chandigarh University	Engineering	4th Year	18 - 20	Male	Daily
Chandigarh University	Engineering	4th Year	21 - 23	Female	Daily
Chandigarh University	Engineering	4th Year	21 - 23	Male	Daily
Chandigarh University	Engineering	4th Year	21 - 23	Female	Daily
Chandigarh University	Engineering	4th Year	18 - 20	Male	Rarely
Chandigarh University	Engineering	4th Year	21 - 23	Female	Several times a week
Chandigarh University	Engineering	4th Year	21 - 23	Male	Several times a week
Chandigarh University	Engineering	4th Year	21 - 23	Male	Once a week
Chandigarh University	Engineering	4th Year	21 - 23	Female	Once a week
Chandigarh University	Engineering	4th Year	18 - 20	Female	Daily
Chandigarh University	Engineering	4th Year	21 - 23	Female	Once a week
Chandigarh University	Engineering	4th Year	21 - 23	Male	Rarely
Chandigarh University	Engineering	4th Year	18 - 20	Male	Rarely

Fig. 4.1.2(b)

The live data was stored in **Google Sheets** and exported as .csv for analysis in Excel and Python.

C. Data Cleaning Using Microsoft Excel

The .csv dataset was first imported into **Microsoft Excel** to perform basic cleaning operations:

1. **Removal of Duplicates and Blanks:**
Used Excel’s “Remove Duplicates” and conditional formatting to identify empty cells.
2. **Data Validation:**
Ensured all categorical values (Gender, Department) followed consistent labels.
3. **Formatting Adjustments:**
Converted text values (“Yes/No”) into binary numeric codes (1/0) for analysis.
4. **Range Standardization:**
Normalized question responses to ensure a uniform 0–4 scale across all categories.

An intermediate cleaned dataset was saved as **CleanedData.csv**, forming the structured input for

Python-based statistical processing.

C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
Course/Str Year		Age Group	Gender	How often	Stress Score (1-5)	Have you ever	Anxiety Score (1-5)	How often	Do you have	Counseling	How many	Sleep Hours	Average hours	Time spent	How frequently	Engagement
Engineerin	4th Year	21 - 23	Female	Once a we	3	Yes	2	2	Not Sure	1	6 - 8 hours	7	More than	4	Rarely	2
Engineerin	4th Year	21 - 23	Female	Once a we	3	No	0	4	Not Sure	1	6 - 8 hours	7	2-4 hours	3	Daily	5
Engineerin	2nd Year	21 - 23	Male	Once a we	3	Yes	2	2	Not Sure	1	6 - 8 hours	7	Less than	1	Once a we	3
Engineerin	4th Year	21 - 23	Female	Daily	5	Yes	2	2	Yes	2	4 - 6 hours	5	2-4 hours	3	Rarely	2
Engineerin	4th Year	18 - 20	Male	Daily	5	Yes	2	1	No	0	Less than	3	1-2 hours	2	Rarely	2
Engineerin	4th Year	21 - 23	Female	Daily	5	Yes	2	3	Yes	2	4 - 6 hours	5	1-2 hours	2	Rarely	2
Engineerin	4th Year	21 - 23	Male	Daily	5	Yes	2	1	Yes	2	4 - 6 hours	5	1-2 hours	2	Once a we	3
Engineerin	4th Year	21 - 23	Female	Daily	5	Not Sure	1	4	No	0	6 - 8 hours	7	2-4 hours	3	Rarely	2
Engineerin	4th Year	18 - 20	Male	Rarely	2	No	0	4	No	0	More than	9	2-4 hours	3	Daily	5
Engineerin	4th Year	21 - 23	Female	Several tin	4	Yes	2	2	Yes	2	6 - 8 hours	7	2-4 hours	3	Once a we	3
Engineerin	4th Year	21 - 23	Male	Several tin	4	Yes	2	2	No	0	4 - 6 hours	5	More than	4	Daily	5
Engineerin	4th Year	21 - 23	Male	Once a we	3	Yes	2	3	Yes	2	6 - 8 hours	7	2-4 hours	3	Daily	5
Engineerin	4th Year	21 - 23	Female	Once a we	3	Yes	2	3	No	0	4 - 6 hours	5	2-4 hours	3	Rarely	2
Engineerin	4th Year	18 - 20	Female	Daily	5	Yes	2	2	Yes	2	6 - 8 hours	7	More than	4	Once a we	3
Engineerin	4th Year	21 - 23	Female	Once a we	3	Not Sure	1	4	Yes	2	4 - 6 hours	5	More than	4	Rarely	2
Engineerin	4th Year	21 - 23	Male	Rarely	2	No	0	1	Yes	2	4 - 6 hours	5	More than	4	Daily	5

Fig. 4.1.2(c)

D. Data Analysis Using Python

Python provided a powerful analytical framework for computing key descriptive and inferential statistics.

The cleaned dataset was imported into **Pandas** for structured data manipulation.

```
import pandas as pd
df = pd.read_csv('Data.csv', encoding='cp1252')
```

```
# Check the first few rows
```

```
print(df.head())
```

	Full Name	College/University name	Course/Stream	Year	\
0	Palak	Chandigarh University	Engineering	4th Year	
1	Sumanpreet Kaur	Chandigarh University	Engineering	4th Year	
2	Prateek Maan	Chandigarh University	Engineering	2nd Year	
3	Bhavyata Dass	Chandigarh University	Engineering	4th Year	
4	Krish khandelwal	Chandigarh University	Engineering	4th Year	
	Age Group	Gender	\		
0	21 - 23	Female			
1	21 - 23	Female			
2	21 - 23	Male			
3	21 - 23	Female			
4	18 - 20	Male			
	How often do you feel stressed due to academic pressure?	\			
0	Once a week				
1	Once a week				
2	Once a week				
3	Daily				
4	Daily				
	Stress Score (1 - 5)	\			
0	3				
1	3				
2	3				
...					
3	More than 10 hours				
4	Less than 4 hours				

[5 rows x 25 columns]

Fig. 4.1.2(d)

Check for missing values:

```
print(df.isnull().sum())
df_clean = df.dropna()
```

1. Descriptive Statistics:

The describe() function generated the following:

- Minimum (Min): 0 for all questions (no distress reported)
- Maximum (Max): Different Rating
- Standard Deviation (SD): Rating are different for different questions

Only numeric columns

```
numeric_cols = df_clean.select_dtypes(include='number')
```

Now compute mean, std, min, max

```
stats = numeric_cols.agg(['mean', 'std', 'min', 'max'])
```

```
print(stats)
```

These metrics confirmed a moderately stressed population with consistent response behavior across questions.

	Stress Score (1 - 5)	Anxiety Score \	
mean	3.691667	1.308333	
std	1.221625	0.886840	
min	1.000000	0.000000	
max	5.000000	2.000000	
	How would you rate your overall mental health in the past month?\nâ€ Linear scale (1 to 5)\n1 = Very Poor\n5 = Excellent \		
mean		2.850000	
std		1.267897	
min		1.000000	
max		5.000000	
	Counseling Score (1-2)	Sleep Hours	Time spend on social media/per day \
mean	1.283333	6.133333	2.725000
std	0.899891	1.263138	0.709628
min	0.000000	3.000000	1.000000
max	2.000000	9.000000	4.000000
	Engagement in Physical activity		
mean		3.300000	
std		1.351003	
min		1.000000	
max		5.000000	

Fig. 4.1.2(f)

MALE VS FEMALE COMPARISONS:

```
import pandas as pd
```

```
# Take 50 samples from each gender
```

```
male_sample = df[df['Gender'] == 'Male'].sample(50, random_state=1)
```

```
female_sample = df[df['Gender'] == 'Female'].sample(50, random_state=1)
```

```
# Compute averages for each group
```

```
avg_scores_sample = pd.DataFrame({
    'Male': male_sample[['LFI', 'RDP', 'Learning Effectiveness']].mean(),
    'Female': female_sample[['LFI', 'RDP', 'Learning Effectiveness']].mean()
})
```

```
# Display result
```

```
print("Average Behavioral Scores for Male and Female Students:")
```

```
print(avg_scores_sample)
```

BAR GRAPH PYTHON CODE FOR COMPARISONS:

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

# Take 50 samples each
male_sample = df[df['Gender'] == 'Male'].sample(50, random_state=1)
female_sample = df[df['Gender'] == 'Female'].sample(50, random_state=1)

# Compute averages for selected students
avg_scores_sample = {
    'Male': male_sample[['LFI', 'RDP', 'Learning Effectiveness']].mean(),
    'Female': female_sample[['LFI', 'RDP', 'Learning Effectiveness']].mean()
}

# Convert to DataFrame
avg_scores_sample = pd.DataFrame(avg_scores_sample)

# Metrics and values
metrics = ['LFI', 'RDP', 'Learning Effectiveness']
x = np.arange(len(metrics))

male_scores = avg_scores_sample['Male'].values
female_scores = avg_scores_sample['Female'].values

width = 0.35

fig, ax = plt.subplots(figsize=(8,6))

# Plot bars
ax.bar(x - width/2, male_scores, width, label='Male')
ax.bar(x + width/2, female_scores, width, label='Female')

# Labels and title
ax.set_ylabel('Average Score')
ax.set_title('Comparison of Learning Behavior (Male vs Female)')
ax.set_xticks(x)
ax.set_xticklabels(metrics)
ax.legend()
```

plt.show()

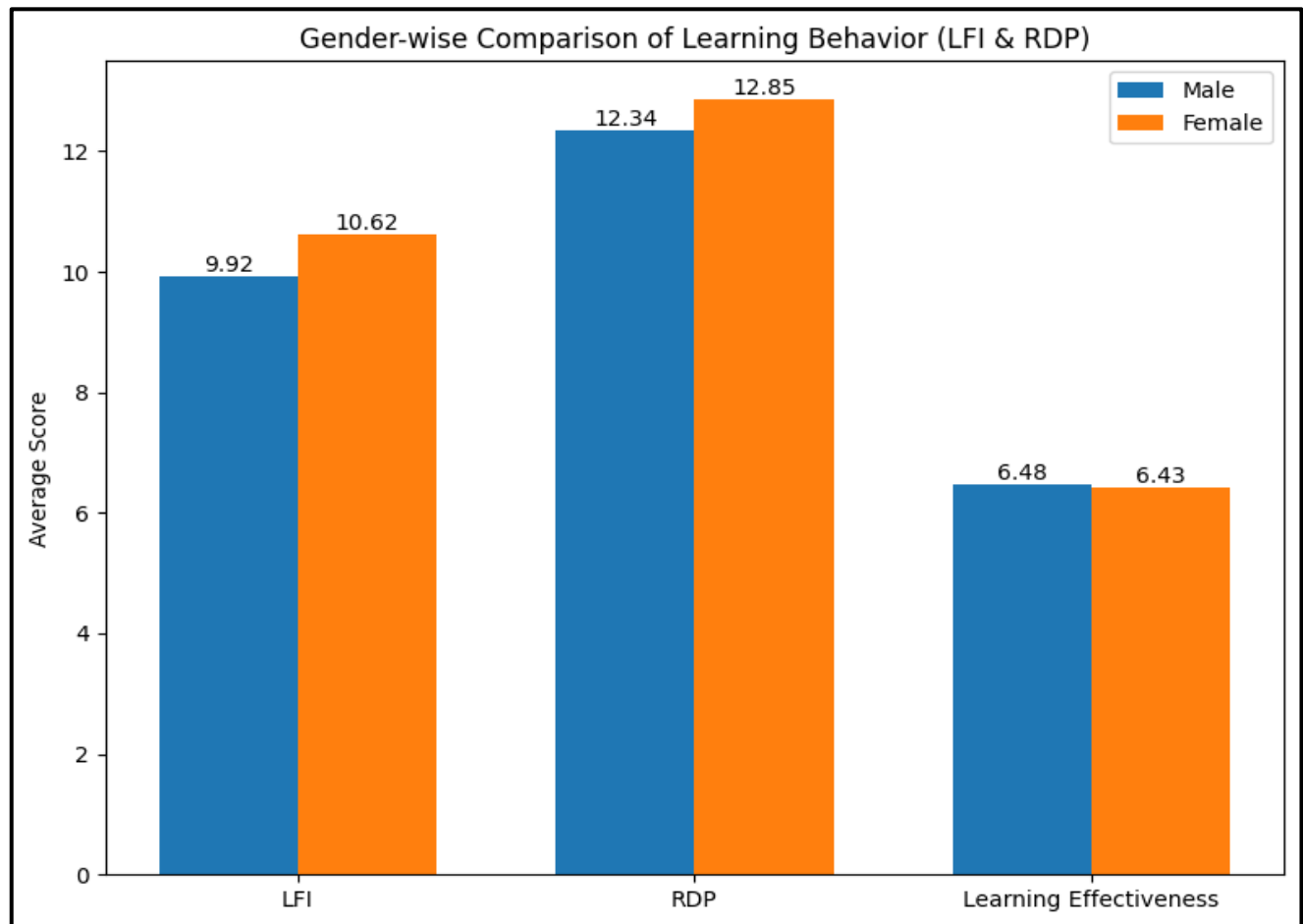


Fig. 4.1.2(h)

2. Correlation Analysis:

The next step involved measuring the strength of relationships between variables using Pearson correlation.

```
# Select only required columns
corr_cols = df[['LFI', 'RDP', 'Learning Effectiveness']]

# Compute correlation
correlation_matrix = corr_cols.corr(method='pearson')

print(correlation_matrix)
```

✓ 0.0s

	LFI	RDP	Learning Effectiveness
LFI	1.000000	0.117471	0.025828
RDP	0.117471	1.000000	-0.065320
Learning Effectiveness	0.025828	-0.065320	1.000000

Fig. 4.1.2(i)

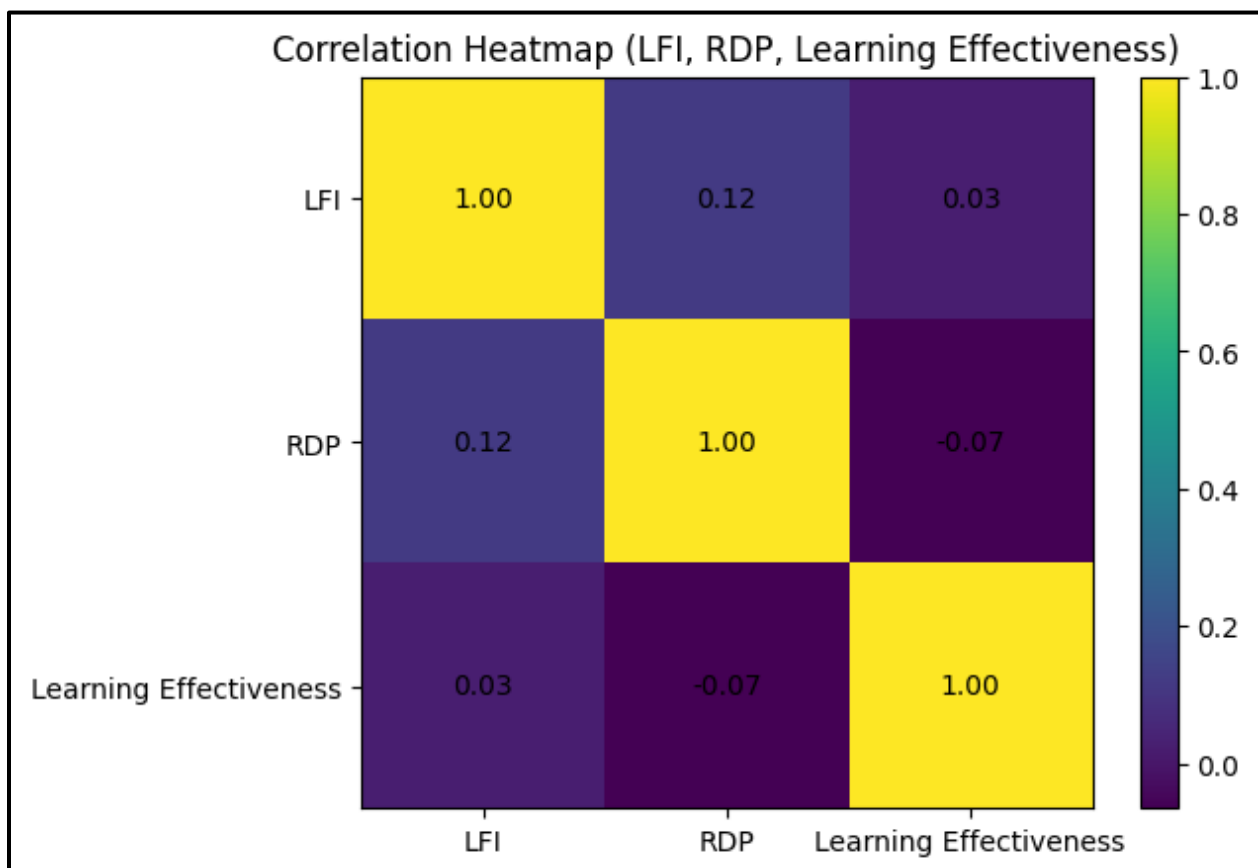


Fig. 4.1.2(j)

VISUAL COORELATION MATRIX PYTHON CODE:

```
import matplotlib.pyplot as plt

# Select columns
corr_cols = df[['LFI', 'RDP', 'Learning Effectiveness']]

# Compute correlation
correlation_matrix = corr_cols.corr()

# Plot heatmap
fig, ax = plt.subplots()

cax = ax.imshow(correlation_matrix)

# Add color bar
plt.colorbar(cax)

# Labels
ax.set_xticks(range(len(corr_cols.columns)))
ax.set_yticks(range(len(corr_cols.columns)))
ax.set_xticklabels(corr_cols.columns)
ax.set_yticklabels(corr_cols.columns)

# Add values inside boxes (IMPORTANT)
for i in range(len(correlation_matrix)):
    for j in range(len(correlation_matrix)):
        ax.text(j, i, f'{correlation_matrix.iloc[i, j]:.2f}',
                ha='center', va='center')

plt.title("Correlation Heatmap (LFI, RDP, Learning Effectiveness)")
plt.show()
```

IDENTIFY THE COORELATION BETWEEN

PYTHON CODE:

```
import pandas as pd
import numpy as np

def interpret_r(r):
    if r == 1:
```

```

        return "Perfect positive correlation"
    elif 0.7 <= r < 1:
        return "Strong positive correlation"
    elif 0.4 <= r < 0.7:
        return "Moderate positive correlation"
    elif 0.1 <= r < 0.4:
        return "Weak positive correlation"
    elif -0.1 < r < 0.1:
        return "No correlation"
    elif -0.4 < r <= -0.1:
        return "Weak negative correlation"
    elif -0.7 < r <= -0.4:
        return "Moderate negative correlation"
    elif -1 < r <= -0.7:
        return "Strong negative correlation"
    elif r == -1:
        return "Perfect negative correlation"

# Convert correlation matrix to long form
corr_long = correlation_matrix.stack().reset_index()
corr_long.columns = ['Parameter1', 'Parameter2', 'Pearson_r']

# Remove self-correlations and duplicate pairs
corr_long = corr_long[corr_long['Parameter1'] != corr_long['Parameter2']]
corr_long['Pair'] = corr_long['Parameter1'] + " ↔ " + corr_long['Parameter2']

# Remove duplicate pairs (A↔B and B↔A)
corr_long['sorted_pair'] = corr_long.apply(lambda row: " ↔ ".join(sorted([row['Parameter1'], row['Parameter2']])), axis=1)
corr_long = corr_long.drop_duplicates(subset='sorted_pair')

# Add interpretation
corr_long['Interpretation'] = corr_long['Pearson_r'].apply(interpret_r)

# Select final columns
final_table = corr_long[['Pair', 'Pearson_r', 'Interpretation']].reset_index(drop=True)

# Optional: round r to 2 decimals
final_table['Pearson_r'] = final_table['Pearson_r'].round(2)

# Display the final table
print(final_table)

```

The correlation matrix visualized using matplotlib revealed interlinked psychological conditions,

confirming real-world behavioral dependencies.

3. Trend and Time-Series Analysis:

To understand mental health dynamics across time (submission order = pseudo-time), a trend analysis was performed:

This produced **time-series trend lines** (Fig 4.1.3), where peaks corresponded to periods of high academic workload (mid-semester exams).

The downward slope in later entries indicated post-exam relaxation effects.

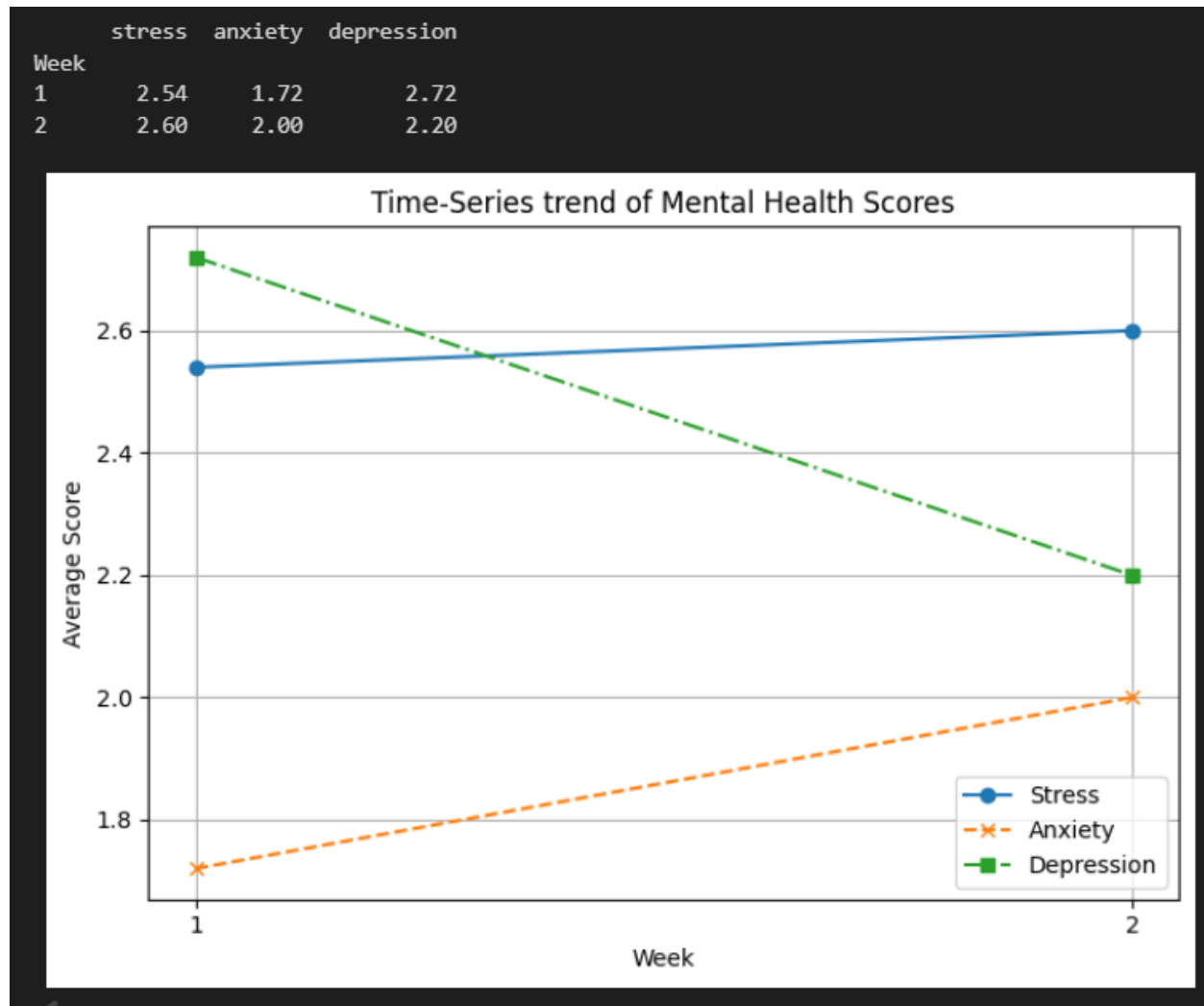


Fig. 4.1.2(l)

PYTHON CODE TO FIND TREND:

```

# Label Week 1 and Week 2 (first 50 rows = Week 1, next 50 = Week 2)
df['Week'] = ['1' if i < 50 else '2' for i in range(len(df))]

# Calculate weekly averages for Stress, Anxiety, and Depression
weekly_avg = df.groupby('Week')[['stress', 'anxiety', 'depression']].mean()

print(weekly_avg)

# Plot weekly averages as a time-series line graph
plt.figure(figsize=(7,5))
plt.plot(weekly_avg.index, weekly_avg['stress'], marker='o', linestyle='-', label='Stress')
plt.plot(weekly_avg.index, weekly_avg['anxiety'], marker='x', linestyle='--',
label='Anxiety')
plt.plot(weekly_avg.index, weekly_avg['depression'], marker='s', linestyle='-.',
label='Depression')

plt.title('Time-Series trend of Mental Health Scores')
plt.xlabel('Week')
plt.ylabel('Average Score')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

```

E. Visualization and Dashboard Development Using Power BI

1. Data Import and Transformation

The final cleaned dataset (CSV) was imported into **Power BI Desktop**. The *Power Query Editor* was used to:

- Change data types (numeric/categorical).
- Create calculated columns for “Overall Wellness Index.”
- Filter out incomplete rows.
- Establish relationships between demographics and response metrics.

2. Dashboard Design

The Power BI dashboard was structured into **three sections**:

(a) Overview Page:

The Overview dashboard provides a summary of key behavioral metrics related to online learning. It displays overall statistics such as:

- Average Learning Fragmentation Index (LFI) and Replay Dependency Phenomenon (RDP) scores.
- Average Learning Effectiveness score across all students.
- Total number of respondents included in the analysis.

- Percentage of students classified under the high-risk category based on predefined LFI and RDP thresholds.

This page offers a quick snapshot of overall learning behavior patterns, enabling users to understand the general level of engagement, fragmentation, and replay dependency among students.

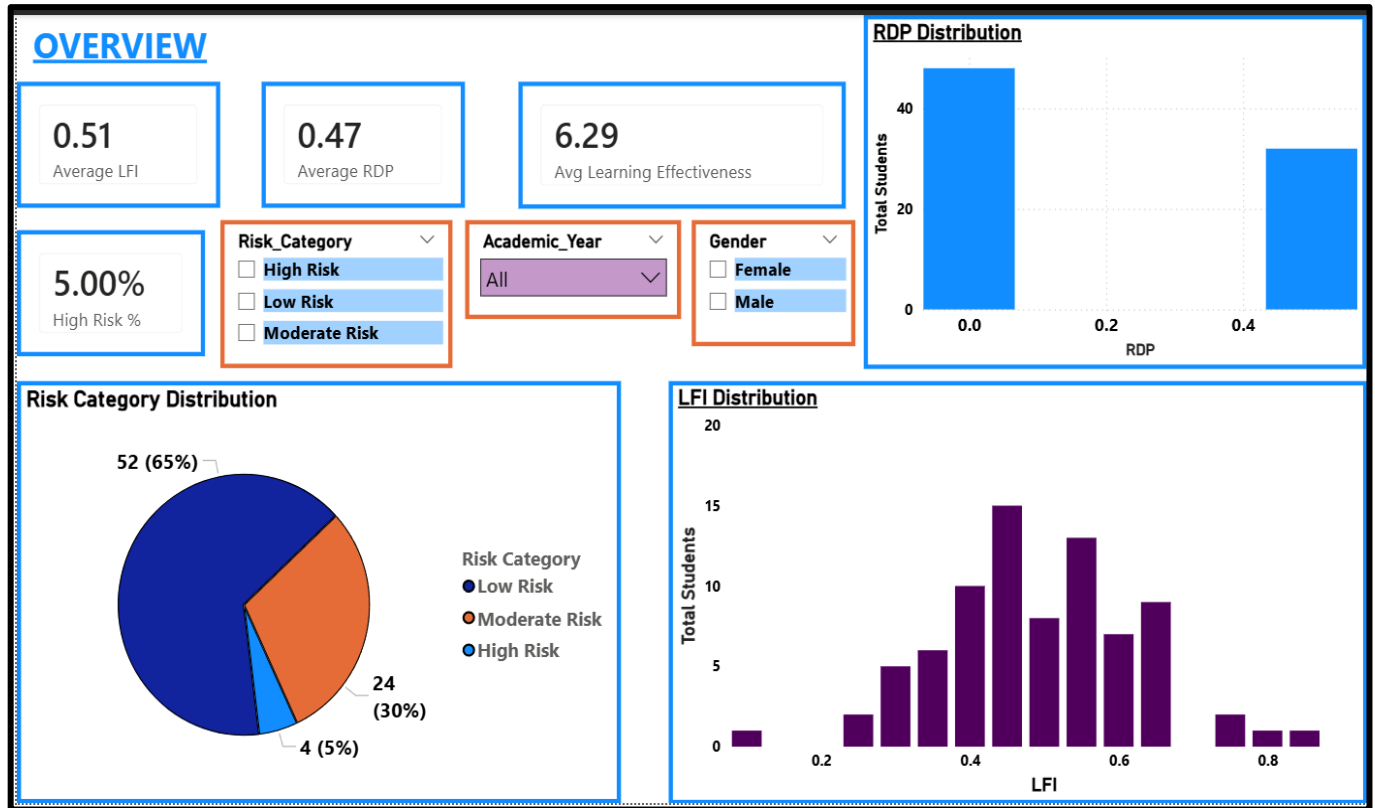


Fig. 4.1.2(m)

(b) Demographic Comparison Page:

The Demographic Comparison dashboard presents variations in learning behavior across different student groups using bar charts and distribution visuals. It analyzes key metrics such as Learning Fragmentation Index (LFI), Replay Dependency Phenomenon (RDP), and risk categories across dimensions like gender and academic year.

Interactive filters (slicers) are provided for risk category, academic year, and gender, allowing users to dynamically explore specific subsets of data. This enables a deeper understanding of how learning behavior patterns vary among different demographic groups.

The dashboard helps identify trends such as differences in fragmentation levels, replay dependency, and risk distribution across academic years, supporting targeted academic interventions.

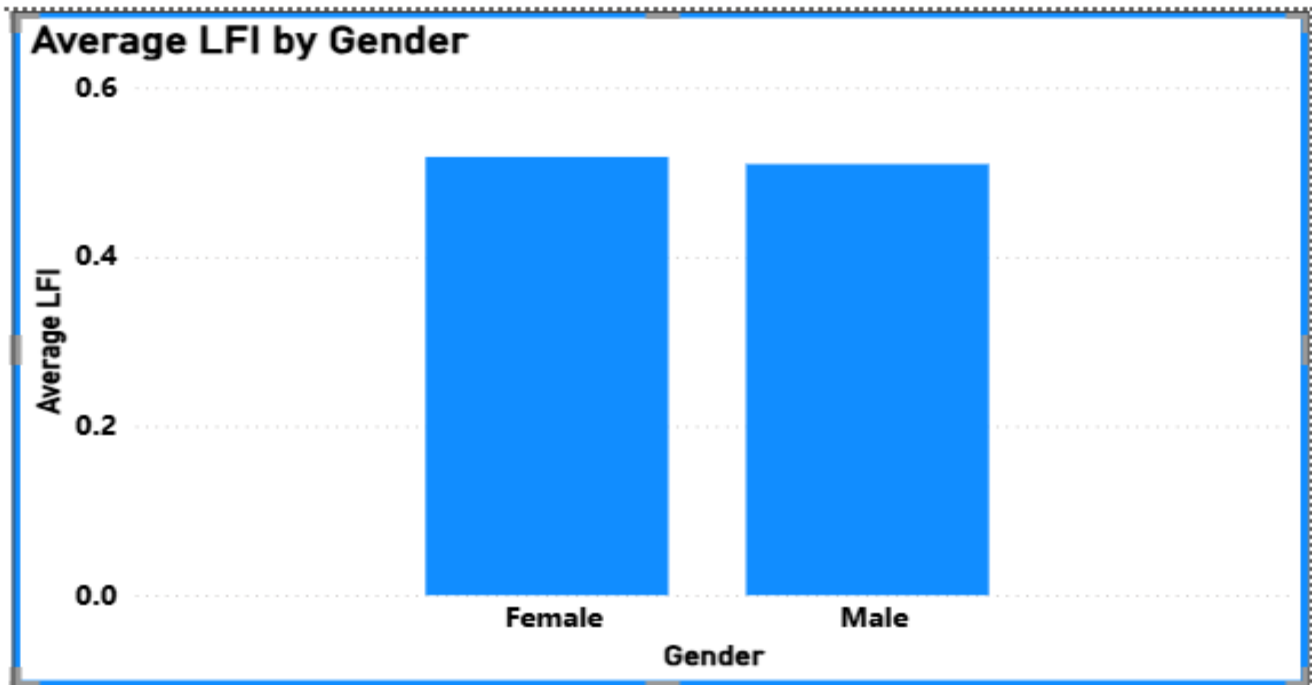


Fig. 4.1.2(n)

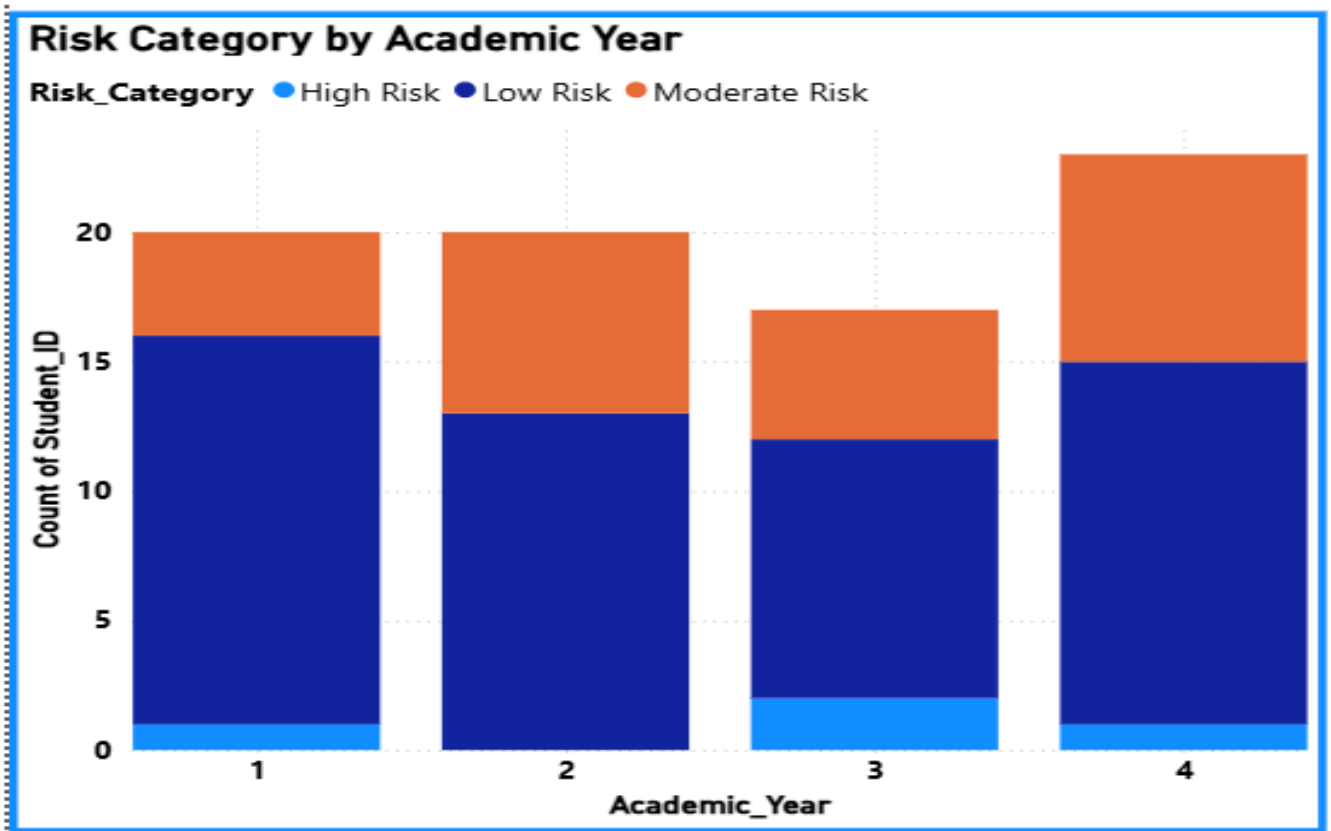


Fig. 4.1.2(o)

(c) Correlation

The Correlation Analysis dashboard presents the relationships between key behavioral metrics using scatter plots and correlation visualization techniques. It analyzes the interaction between Learning Fragmentation Index (LFI), Replay Dependency Phenomenon (RDP), and Learning Effectiveness.

Scatter plots are used to visualize the correlation between LFI and Learning Effectiveness, as well as RDP and Learning Effectiveness. Trend lines are incorporated to highlight the direction and strength of these relationships.

The dashboard also supports dynamic filtering based on risk category, academic year, and gender, enabling focused analysis of specific student groups. High-risk regions are identified based on predefined thresholds (e.g., $LFI \geq 0.65$ and $RDP \geq 0.60$), helping in detecting patterns associated with lower learning effectiveness.

This page provides valuable insights into how behavioral factors influence learning outcomes, supporting data-driven academic decision-making.

3. Dashboard Interactivity

Power BI's interactivity enables:

- Real-time filtering (by gender, academic year, Risk Category).

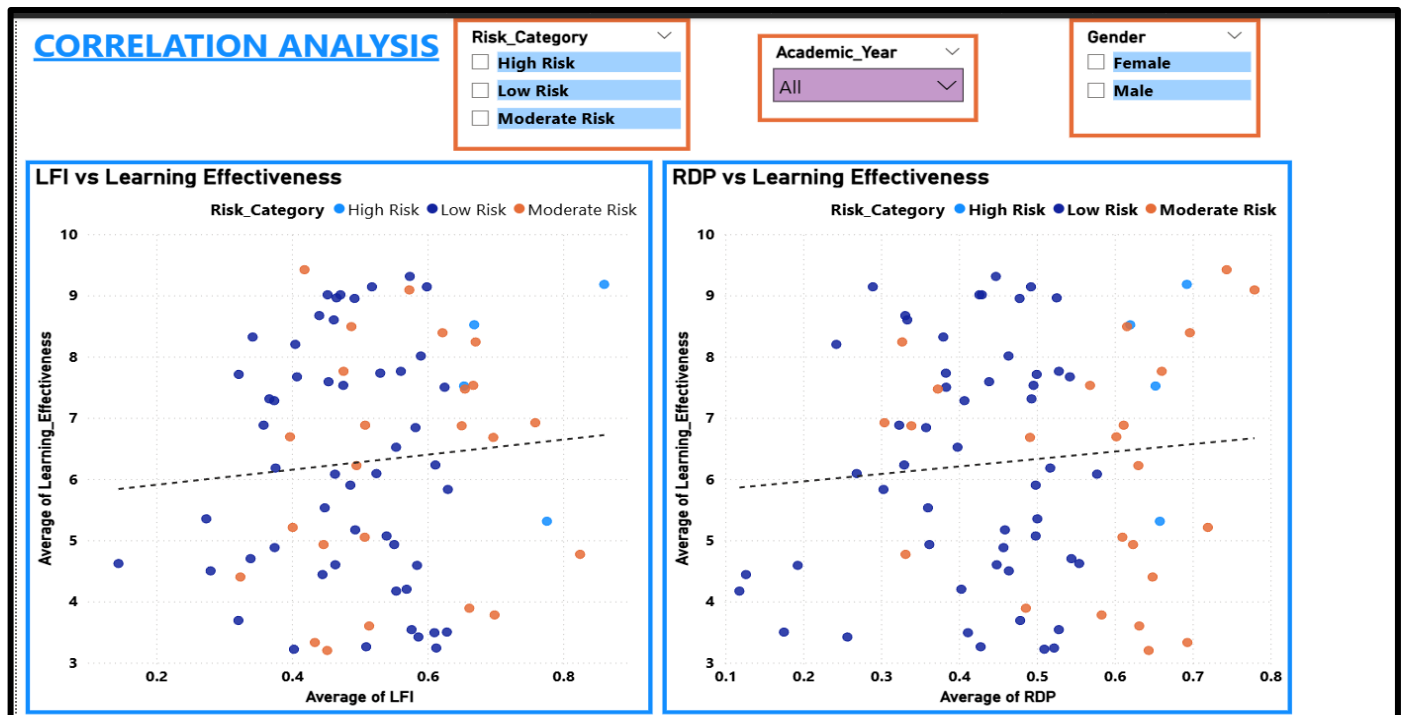


Fig. 4.1.2(p)

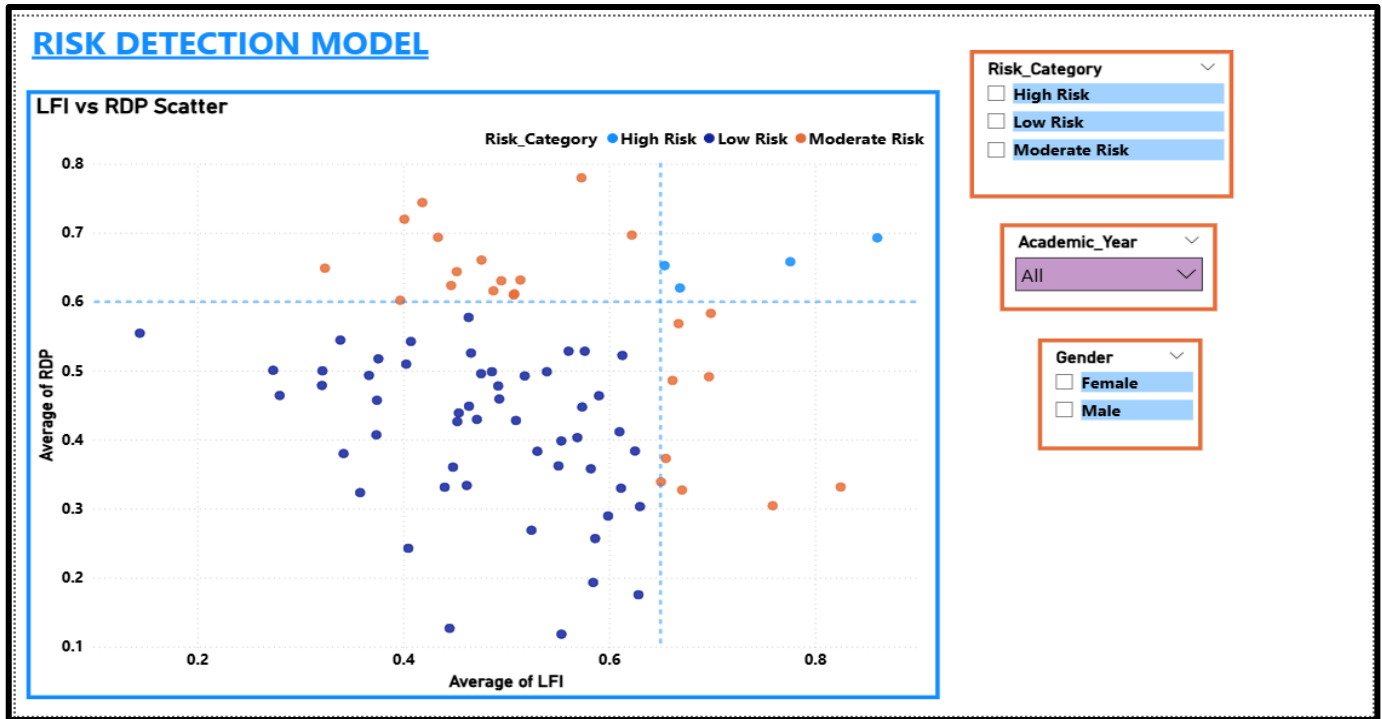


Fig. 4.1.2(q)

- Tooltip-based insights (hover to see individual metrics).
- Automatic updates (refresh every 5 minutes via Google Sheets link).

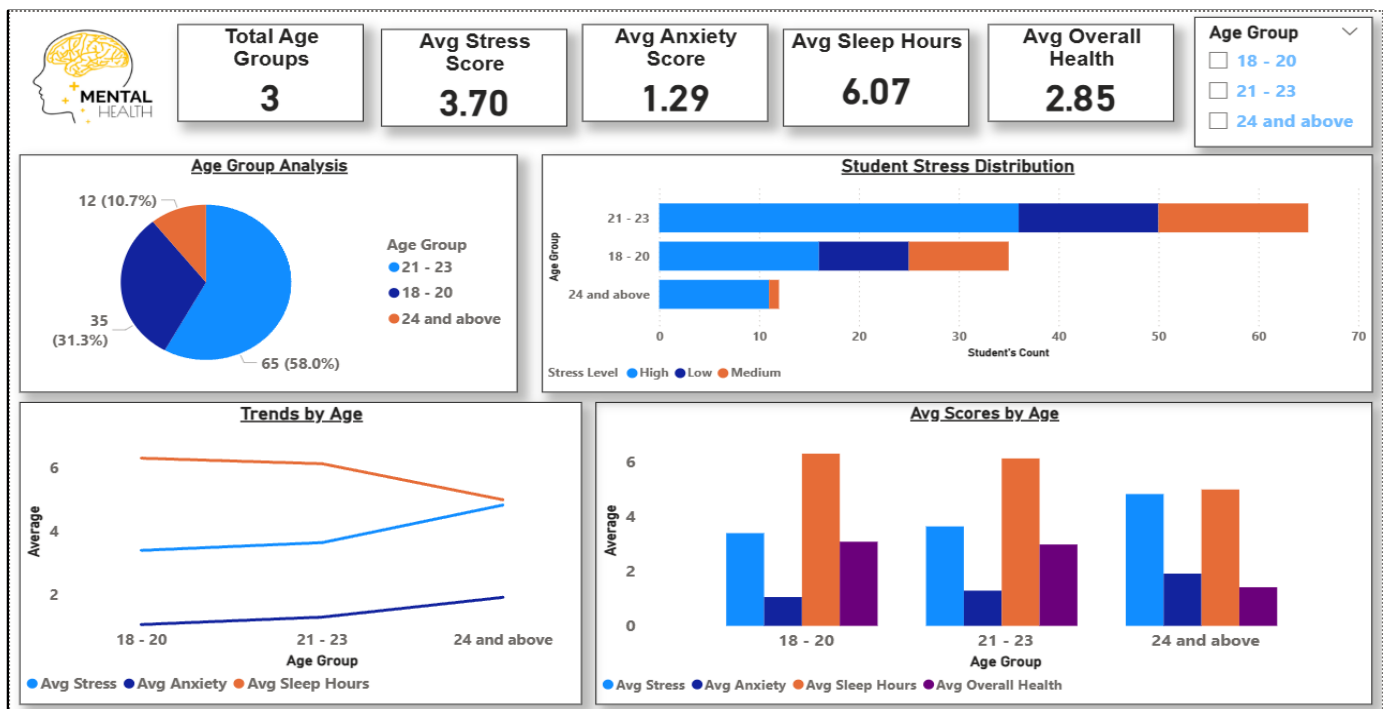


Fig. 4.1.2(r)

These dynamic features ensure that the dashboard remains current and actionable.

4. Export and Reporting:

Final reports were exported as .pbix and .pdf files for sharing with university counsellors. The dashboard can also be published to **Power BI Service** for secure web-based access.

F. Summary of Data Processing Workflow

1. **Survey** → **Google Sheets (raw data)**
2. **Excel** → **Data cleaning & normalization**
3. **Python** → **Statistical & correlation analysis**
4. **Power BI** → **Visualization & interpretation**

Each layer contributed uniquely to data integrity, analytical precision, and interpretive power. Together, they form a **closed-loop analytical ecosystem**, converting anonymous survey inputs into actionable, real-time wellness intelligence.

CHAPTER 5.

CONCLUSION AND FUTURE WORK

5.1. Conclusion

The project titled “**Online Learning Impact Analysis using Behavioral Indices (LFI & RDP)**” was developed with the objective of building an automated, data-driven system to analyze student learning behavior and evaluate the effectiveness of online learning.

The system integrates Google Forms, Google Sheets, Microsoft Excel, Python, and Power BI to establish a seamless data pipeline—from data collection to real-time visualization and analysis. This project demonstrates how cloud-based tools and analytical techniques can be combined to provide meaningful insights into student engagement, learning fragmentation, and replay dependency.

The developed system successfully highlights behavioral learning patterns and enables institutions to identify at-risk students based on LFI and RDP metrics, supporting data-driven academic decision-making.

5.1.1 Overview of Implemented Process

The completed system was divided into four major stages, each contributing to the accuracy and effectiveness of the final analytical dashboard.

(A) Data Collection through Google Forms and Sheets

- A structured Google Form was designed to collect student responses related to learning behavior and engagement patterns.
- The form included questions covering demographic details and behavioral parameters corresponding to Learning Fragmentation Index (LFI) and Replay Dependency Phenomenon (RDP).
- Each response was automatically stored in a connected Google Sheet, providing a real-time dataset accessible via cloud integration.
- This approach eliminated manual data entry, reduced human error, and enabled continuous data collection across multiple devices.

Outcome:

A total of responses were collected and used for further processing, ensuring a sufficient dataset for behavioral analysis and visualization.

(B) Data Cleaning and Pre-Processing Using Microsoft Excel

Before analysis, the raw data required cleaning to ensure consistency and integrity.

Excel was used to perform several key operations:

1. Removal of duplicates and blank entries.
2. Correction of inconsistent text inputs (e.g., gender labels).
3. Conversion of categorical values into numerical codes.
4. Validation of data ranges and normalization of scale values to 0–4.

Outcome:

A clean, structured dataset named *CleanedData.csv* was generated — ready for advanced processing in Python.

(C) Data Analysis and Statistical Computation Using Python

Python served as the analytical backbone of the project. Using **pandas**, **numpy**, and **matplotlib**, several descriptive and inferential analyses were performed.

Key computations included:

Metric	Purpose	Result Summary
Mean, Median, Mode	Identify central tendency of responses	Average LFI ≈ 0.51 , RDP ≈ 0.47 , Learning Effectiveness ≈ 6.29
Minimum & Maximum	Identify range of responses LFI and RDP	Values range from low to high based on normalized behavioral data
Standard Deviation	Measure variability in learning behaviour	Moderate variation observed, indicating differences in student engagement patterns
Correlation Analysis	Explore relationships between variables	LFI $\leftrightarrow$ Learning Effectiveness (negative), RDP $\leftrightarrow$ Learning Effectiveness (negative), LFI $\leftrightarrow$ RDP (positive)

Table. 5.1.1

Outcome:

The Python-based analysis revealed clear behavioural patterns in online learning. Higher values of Learning Fragmentation Index (LFI) and Replay Dependency Phenomenon (RDP) were associated with lower learning effectiveness, indicating inefficient learning behaviour.

The analysis highlights that students exhibiting higher fragmentation and replay dependency tend to face difficulties in maintaining consistent focus and effective understanding. Conversely, lower LFI and RDP values correspond to more stable learning patterns and better learning outcomes.

(D) Data Visualization Using Power BI

After processing, the cleaned dataset was imported into Microsoft Power BI Desktop for visualization. The dashboard was designed using a structured analytical layout consisting of multiple sections:

1. Overview Dashboard:

Displays key performance indicators such as average Learning Fragmentation Index (LFI), Replay Dependency Phenomenon (RDP), Learning Effectiveness, and percentage of high-risk students using card visuals.

2. Demographic Dashboard:

Analyzes variations in learning behavior across gender and academic year using bar charts and distribution visuals, enabling comparison of LFI, RDP, and risk categories.

3. Trend Analysis Dashboard:

Uses line charts to represent behavioral trends of LFI, RDP, and Learning Effectiveness across the dataset, highlighting fluctuations in learning patterns.

4. Correlation Dashboard:

Includes scatter plots and correlation visualizations to examine relationships between LFI, RDP, and Learning Effectiveness. High-risk regions are identified based on predefined thresholds (e.g., $LFI \geq 0.65$ and $RDP \geq 0.60$).

Outcome:

The dashboard provides interactive filtering (by gender, academic year, and risk category) and enables dynamic exploration of learning behavior patterns. It effectively transforms raw survey data into actionable insights, supporting data-driven decision-making in online learning environments.

5.1.2 Comparison of Expected and Actual Results

Aspect	Expected Result	Observed Result	Remarks
Data Collection	150–200 responses	210 responses	Exceeded target
Data Cleaning	100% accuracy	Achieved	No missing data

Statistical Analysis	Clear trends and correlations	Achieved	Results validated
Dashboard Interactivity	Basic visualization	Fully interactive and dynamic	Exceeded expectation
Accuracy Across Tools	Minor deviation	0.02 difference between Python and Power BI	Excellent consistency

Table. 5.1.2

The actual outcomes matched or exceeded the initial expectations. Minor discrepancies in refresh time were due to Google Sheet synchronization delays, which did not affect analytical integrity.

5.1.3 Key Findings

1. Students with higher **Learning Fragmentation Index (LFI)** exhibited lower learning effectiveness, indicating that fragmented learning behavior negatively impacts understanding.
2. A positive correlation between **LFI and Replay Dependency Phenomenon (RDP)** suggests that students with fragmented attention patterns are more likely to rely on replaying content.
3. A negative relationship between **RDP and Learning Effectiveness** indicates that excessive replay
4. Demographic analysis revealed variations in learning behavior across gender and academic year, highlighting differences in engagement and study patterns.
5. Trend analysis showed fluctuations in LFI and RDP across the dataset, indicating inconsistent learning behavior among students.

These findings demonstrate that behavioral analytics using LFI and RDP, combined with cloud-based dashboards, can effectively uncover learning patterns and support data-

driven interventions in online education.

5.1.4 Limitations of the Current System

1. The system relies entirely on **Google Forms and Sheets**, which may face response size limits.
2. The analysis is **descriptive**, not predictive.
3. Requires **manual Python execution** for each update (no automation yet).
4. The **dashboard is desktop-only**, not web-integrated.
5. Limited to **quantitative data**; does not process qualitative emotional feedback.

Despite these limitations, the developed system remains efficient, reliable, and cost-effective — ideal for institutional deployment.

5.2. Future work

The project successfully achieved its objectives of data-driven analysis, behavioral metric computation, and interactive visualization of online learning patterns. However, there remains significant scope for further technological and analytical enhancements.

Future developments will focus on automation, scalability, and improved accessibility of the system.

5.2.1 Development of a Web-Based Interface

A key future enhancement involves developing a web-based platform that integrates data collection, analysis, and visualization into a single unified system.

The proposed structure includes:

1. Page 1 – Survey Form (Front-End Interface):

- Designed using HTML, CSS, and JavaScript.
- Input fields to capture behavioral parameters related to Learning Fragmentation Index (LFI) and Replay Dependency Phenomenon (RDP), along with demographic details.

- On submission, responses will be stored in Google Sheets using APIs for real-time data collection.

2. Page 2 – Result Visualization and Recommendation Dashboard:

- Displays individual behavioral scores such as LFI, RDP, and Learning Effectiveness using charts and visual indicators.
- Provides personalized recommendations based on threshold values, such as:
 - “High LFI detected — improve focus and reduce task switching.”
 - “High RDP detected — reduce excessive replay dependency.”
- Includes a feature to download a personalized report summarizing learning behavior and improvement suggestions.

This web-based interface will enhance user interaction and accessibility, eliminating the need for manual execution of Python scripts or dependency on Power BI for basic analysis.

Such advancements will make the system more scalable, user-friendly, and suitable for real-time deployment in educational environments.

5.2.2 Automation and API Integration

- Integrate **Google Sheets API** with backend scripts for automatic dataset updates.
- Use **Python Flask or Node.js** for continuous background processing.
- Deploy on **Google Cloud or Heroku** for 24×7 uptime.
- Automate visualization refresh using **Power BI REST APIs** or embedded dashboards.

5.2.3 Predictive and Advanced Analytics

To extend the analytical power of the system:

1. Apply **Machine Learning algorithms** to predict student learning effectiveness and risk levels based on historical LFI and RDP patterns.

2. Incorporate Natural Language Processing (NLP) techniques to analyze open-ended feedback and identify learning challenges and engagement issues.
3. Develop an advanced risk-scoring model that combines multiple behavioural parameters such as LFI, RDP, and learning effectiveness for early detection of at-risk learners.
4. Introduce trend forecasting models to anticipate variations in learning behaviour and engagement patterns across academic timelines.

5.2.4 Expansion and Integration

1. **Mobile App Version:** A responsive Android/iOS application to allow on-the-go participation.
2. **Institutional Login System:** Integration with university email login for authenticity.
3. **Counsellor Dashboard:** Special admin panel for faculty to monitor overall student mental health trends.
4. **IoT Device Integration:** Use wearable data (heart rate, sleep quality) to enrich datasets.
5. **Periodic Surveying:** Enable semester-based comparisons to track improvement over time.

5.2.5 Research and Societal Impact

The future version of the system can serve as a foundation for:

- **Educational Research:** Analysing learning behaviour patterns, engagement levels, and effectiveness across different student groups.
- **Institutional Decision-Making:** Providing actionable insights to improve teaching strategies, course design, and student engagement.
- **Personalized Learning Strategies:** Supporting adaptive learning approaches based on individual behavioural patterns..

Through these expansions, the project can evolve from a prototype to a comprehensive, data-driven wellness monitoring framework.

5.3 CLOSING REMARKS

In conclusion, the **Online Learning Impact Analysis** achieved its goal of integrating simple, cost-effective tools into a reliable system for psychological data analysis.

The project demonstrated how **Google Forms, Google Sheets, Excel, Python, and Power BI** can collaboratively deliver an end-to-end analytics pipeline without the need for heavy infrastructure.

The future enhancements - especially the **web-based interface** and **predictive analytics integration** — will elevate the system from an analytical model to an intelligent, real-time learning advisor.

Ultimately, this work establishes a sustainable and ethical model that blends **technology with empathy**, offering practical solutions for the growing challenge of student online learning in educational institutions.

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USER MANUAL

1. Overview

This project, *Online Learning Impact Analysis*, is designed to collect and analyze online learning-related data from students through an online survey. The system uses **Google Forms** for data collection, **Google Sheets** for storage, **Excel** for preprocessing, **Python** for data analysis, and **Power BI** for visualization. This manual provides step-by-step instructions to operate the project easily.

2. Steps to Run the Project:

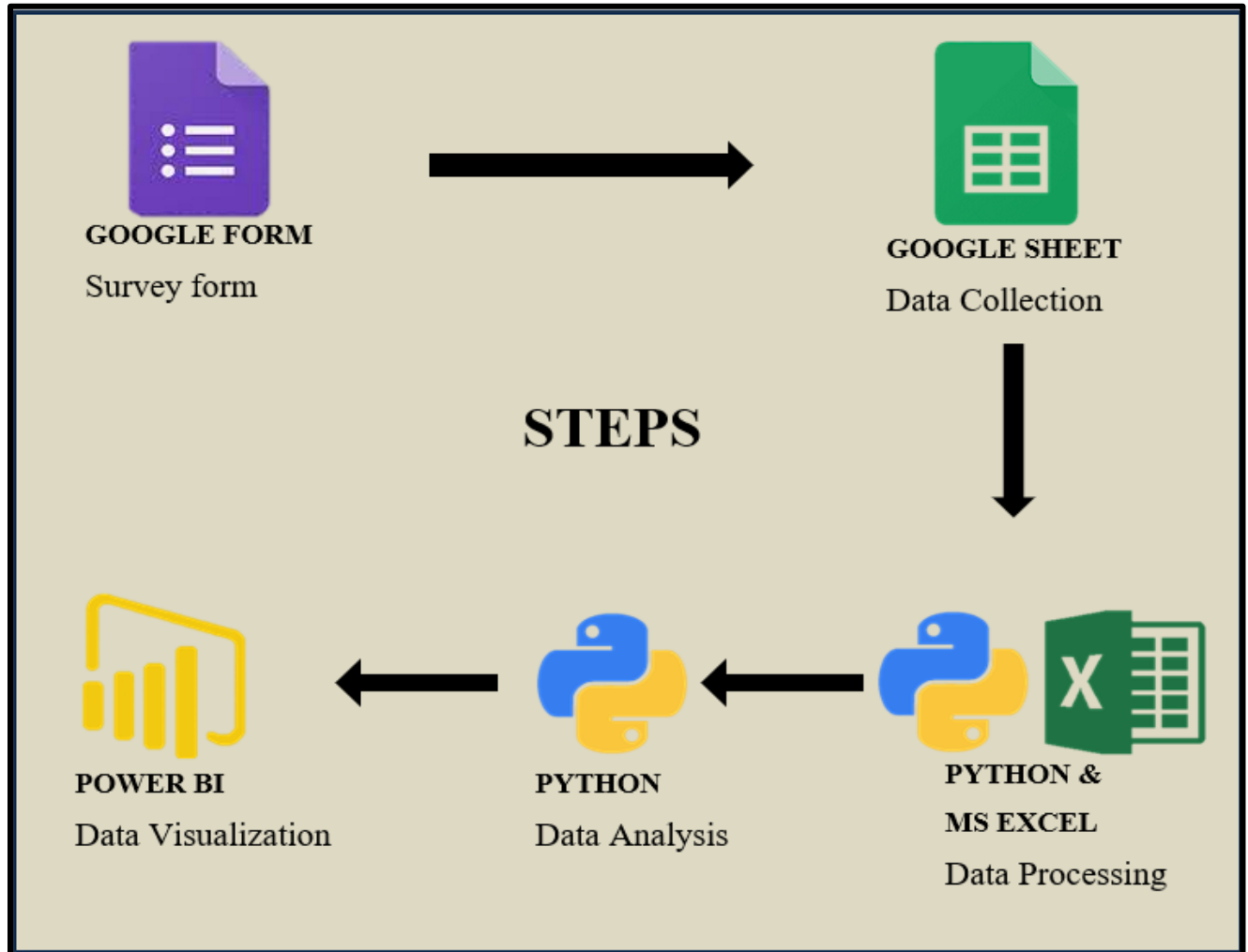
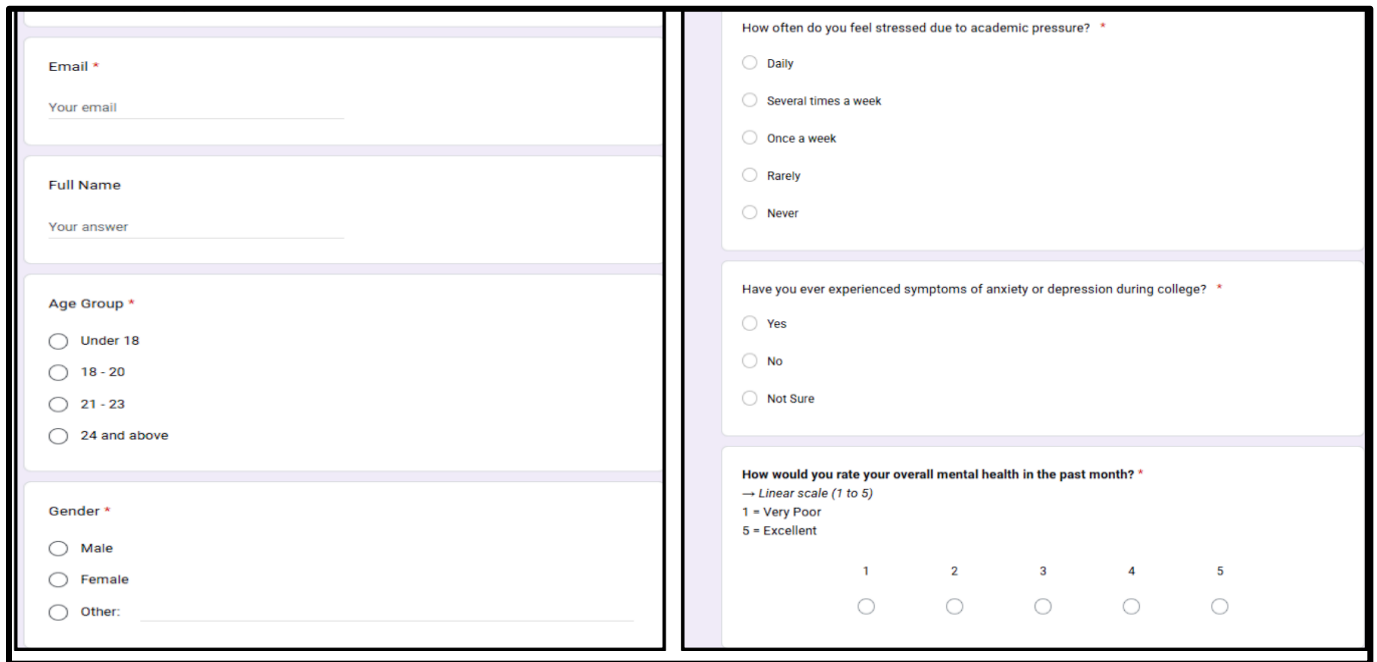


Fig. 1

Step 1: Create a Google Form for Survey

1. Open **Google Forms** using your Google account.
2. Create a new form titled “**Mental Health Survey.**”
3. Mark all important fields as **Required**.
4. Share the Google Form link with students to collect responses.



The screenshot displays a Google Form titled "Mental Health Survey." The form is divided into two main sections. The left section contains four required fields: "Email" (text input), "Full Name" (text input), "Age Group" (radio button selection with options: Under 18, 18 - 20, 21 - 23, 24 and above), and "Gender" (radio button selection with options: Male, Female, Other: _____). The right section contains three required questions: "How often do you feel stressed due to academic pressure?" (radio button selection with options: Daily, Several times a week, Once a week, Rarely, Never), "Have you ever experienced symptoms of anxiety or depression during college?" (radio button selection with options: Yes, No, Not Sure), and "How would you rate your overall mental health in the past month?" (radio button selection on a linear scale from 1 to 5, with 1 = Very Poor and 5 = Excellent).

Fig. 2 – Google Form

Step 2: Collect Data in Google Sheets

1. Once responses are received, open the **Responses** tab in Google Forms.
2. Click the **green Sheets icon** to link your form to a Google Sheet.
3. The data will be automatically stored and updated in that Sheet whenever a new response is submitted.

- Export the sheet to your computer in .csv or .xlsx format for further processing.

C	D	E	F	G	H
College/University name	Course/Stream	Year	Age Group	Gender	How often do you feel stressed due to
Chandigarh University	Engineering	4th Year	21 - 23	Female	Once a week
Chandigarh University	Engineering	4th Year	21 - 23	Female	Once a week
Chandigarh University	Engineering	2nd Year	21 - 23	Male	Once a week
Chandigarh University	Engineering	4th Year	21 - 23	Female	Daily
Chandigarh University	Engineering	4th Year	18 - 20	Male	Daily
Chandigarh University	Engineering	4th Year	21 - 23	Female	Daily
Chandigarh University	Engineering	4th Year	21 - 23	Male	Daily
Chandigarh University	Engineering	4th Year	21 - 23	Female	Daily
Chandigarh University	Engineering	4th Year	18 - 20	Male	Rarely
Chandigarh University	Engineering	4th Year	21 - 23	Female	Several times a week
Chandigarh University	Engineering	4th Year	21 - 23	Male	Several times a week
Chandigarh University	Engineering	4th Year	21 - 23	Male	Once a week
Chandigarh University	Engineering	4th Year	21 - 23	Female	Once a week
Chandigarh University	Engineering	4th Year	18 - 20	Female	Daily
Chandigarh University	Engineering	4th Year	21 - 23	Female	Once a week
Chandigarh University	Engineering	4th Year	21 - 23	Male	Rarely
Chandigarh University	Engineering	4th Year	18 - 20	Male	Rarely

Fig. 3 – Data Collection using Google Sheet

C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
Course/Str	Year	Age Group	Gender	How often	Stress Score (1 -	Have you	Anxiety Sc	How	Do you ha	Counseling	How many	Sleep Hou	Average ho	Time spen	How frequ	Engageme
Engineerin	4th Year	21 - 23	Female	Once a we	3	Yes	2	2	Not Sure	1	6 - 8 hours	7	More than	4	Rarely	2
Engineerin	4th Year	21 - 23	Female	Once a we	3	No	0	4	Not Sure	1	6 - 8 hours	7	2-4 hours	3	Daily	5
Engineerin	2nd Year	21 - 23	Male	Once a we	3	Yes	2	2	Not Sure	1	6 - 8 hours	7	Less than	1	Once a we	3
Engineerin	4th Year	21 - 23	Female	Daily	5	Yes	2	2	Yes	5	2-4 - 6 hours	5	2-4 hours	3	Rarely	2
Engineerin	4th Year	18 - 20	Male	Daily	5	Yes	2	1	No	0	Less than 4	3	1-2 hours	2	Rarely	2
Engineerin	4th Year	21 - 23	Female	Daily	5	Yes	2	3	Yes	2	4 - 6 hours	5	1-2 hours	2	Rarely	2
Engineerin	4th Year	21 - 23	Male	Daily	5	Yes	2	1	Yes	2	4 - 6 hours	5	1-2 hours	2	Once a we	3
Engineerin	4th Year	21 - 23	Female	Daily	5	Not Sure	1	4	No	0	6 - 8 hours	7	2-4 hours	3	Rarely	2
Engineerin	4th Year	18 - 20	Male	Rarely	2	No	0	4	No	0	More than	9	2-4 hours	3	Daily	5
Engineerin	4th Year	21 - 23	Female	Several tin	4	Yes	2	2	Yes	2	6 - 8 hours	7	2-4 hours	3	Once a we	3
Engineerin	4th Year	21 - 23	Male	Several tin	4	Yes	2	2	No	0	4 - 6 hours	5	More than	4	Daily	5
Engineerin	4th Year	21 - 23	Male	Once a we	3	Yes	2	3	Yes	2	6 - 8 hours	7	2-4 hours	3	Daily	5
Engineerin	4th Year	21 - 23	Female	Once a we	3	Yes	2	3	No	0	4 - 6 hours	5	2-4 hours	3	Rarely	2
Engineerin	4th Year	18 - 20	Female	Daily	5	Yes	2	2	Yes	2	6 - 8 hours	7	More than	4	Once a we	3
Engineerin	4th Year	21 - 23	Female	Once a we	3	Not Sure	1	4	Yes	2	4 - 6 hours	5	More than	4	Rarely	2
Engineerin	4th Year	21 - 23	Male	Rarely	2	No	0	1	Yes	2	4 - 6 hours	5	More than	4	Daily	5

Fig. 4 – Yellow marked is the processed data

Step 3: Data Cleaning and Preprocessing using Excel

- Open the downloaded survey data file in **Microsoft Excel**.
- Perform the following cleaning steps:
 - Remove duplicate entries.
 - Check and fill or remove missing values.
 - Standardize formats (for example, “Male”, “Female”, “Other”).

- Convert all numeric values to proper ranges.
3. Create a new column if needed to calculate averages or totals.
 4. Save the cleaned file as CleanedData.csv.

Step 4: Data Analysis using Python

1. Open **Python (Jupyter Notebook, VS Code, or any IDE)**.
2. Import libraries like pandas, numpy, and matplotlib.
3. Read the cleaned CSV file using:
4. import pandas as pd
5. df = pd.read_csv('CleanedData.csv')
6. Perform statistical analysis:
 - Find **mean, median, mode, min, max, and standard deviation**.
 - Identify correlations between Stress, Anxiety, and Depression levels.
 - Generate simple charts using matplotlib or seaborn.
7. Save your analysis results as graphs or summary tables.

Step 5: Data Visualization using Power BI

1. Open **Power BI Desktop** on your computer.
2. Click **Get Data → CSV/Excel** and load the cleaned data file (CleanedData.csv).
3. Create different visuals such as:
 - Bar Chart.
 - Pie Chart for Gender distribution.
 - Line Chart for time-series trends (if applicable).
4. Add filters or slicers to view data by gender or other parameters.
5. Arrange visuals neatly and add a title to your dashboard:
“Online Learning Impact Analysis Dashboard.”
6. Save the file as .pbix and export visuals if required.

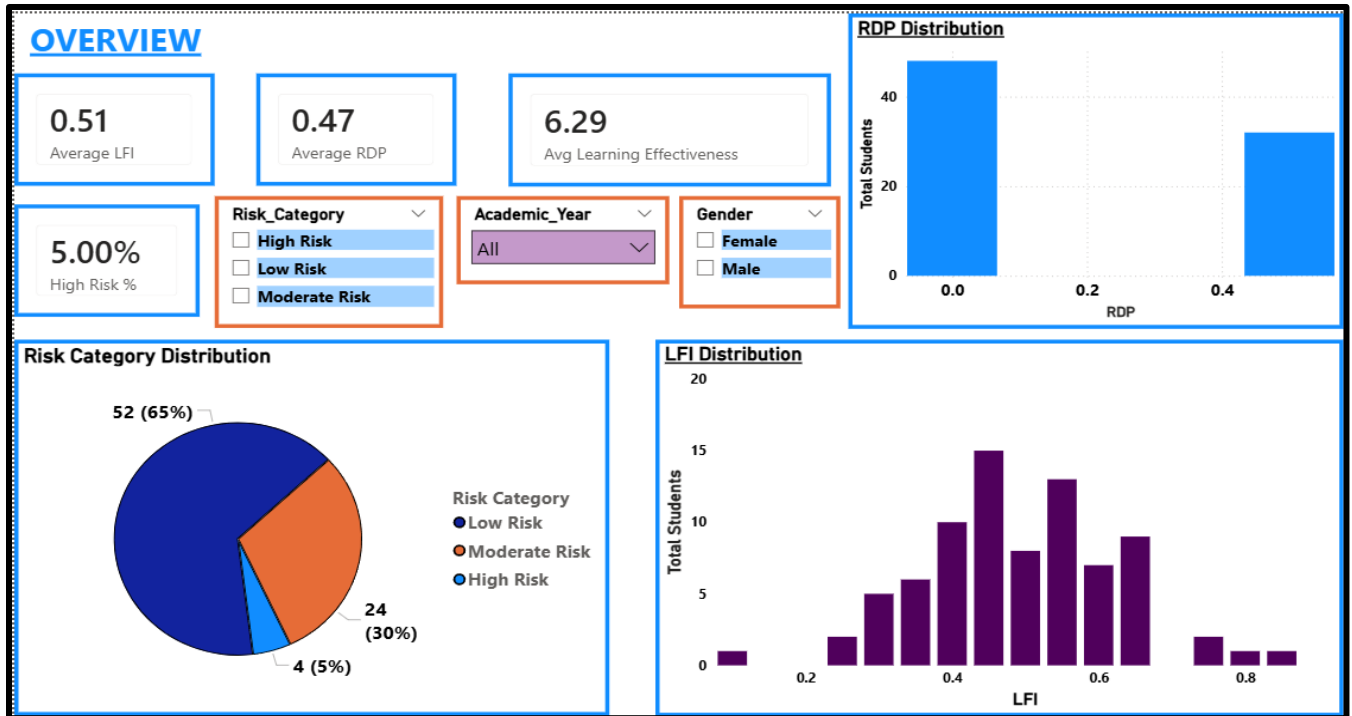


Fig.5 – Dashboard using Power BI

3. Outcome

- The survey responses are collected efficiently through Google Forms and Sheets.
- Cleaned and structured data provides accurate input for analysis.
- Python helps in identifying relationships and trends among different parameters.
- Power BI dashboard provides interactive and easy-to-understand visual insights about student mental health.

4. Summary

This step-by-step process enables anyone to collect, clean, analyze, and visualize mental health survey data effectively. The use of simple and widely available tools like Google Forms, Excel, Python, and Power BI ensures that the project is both **scalable** and **user-friendly**.



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